

Article

Research and Development of an IoT Smart Irrigation System for Farmland Based on LoRa and Edge Computing

Ying Zhang ^{1,2,3,4}, Xingchen Wang ^{1,2,3,4}, Liyong Jin ⁵, Jun Ni ^{1,2,3,4}, Yan Zhu ^{1,2,3,4} , Weixing Cao ^{1,2,3,4} and Xiaoping Jiang ^{1,2,3,4,*} 

- ¹ College of Agriculture, Nanjing Agricultural University, Nanjing 210095, China; 2022101186@stu.njau.edu.cn (Y.Z.); 2022101185@stu.njau.edu.cn (X.W.); nijun@njau.edu.cn (J.N.); yanzhu@njau.edu.cn (Y.Z.); caow@njau.edu.cn (W.C.)
- ² Jiangsu Collaborative Innovation Center for the Technology and Application of Internet of Things, Nanjing 210095, China
- ³ National Information Agricultural Engineering Technology Center, Nanjing 210095, China
- ⁴ Engineering Research Center of Smart Agriculture, Ministry of Education, Nanjing 210095, China
- ⁵ Jiangsu Rainfolds Irrigation System Co., Ltd., Nanjing 210011, China; jly_jris@163.com
- * Correspondence: xpjiang@njau.edu.cn; Tel.: +86-25-8439-6593

Abstract: In response to the current key issues in the field of smart irrigation for farmland, such as the lack of data sources and insufficient integration, a low degree of automation in drive execution and control, and over-reliance on cloud platforms for analyzing and calculating decision making processes, we have developed nodes and gateways for smart irrigation. These developments are based on the EC-IOT edge computing IoT architecture and long range radio (LoRa) communication technology, utilizing STM32 MCU, WH-101-L low-power LoRa modules, 4G modules, high-precision GPS, and other devices. An edge computing analysis and decision model for smart irrigation in farmland has been established by collecting the soil moisture and real-time meteorological information in farmland in a distributed manner, as well as integrating crop growth period and soil properties of field plots. Additionally, a mobile mini-program has been developed using WeChat Developer Tools that interacts with the cloud via the message queuing telemetry transport (MQTT) protocol to realize data visualization on the mobile and web sides and remote precise irrigation control of solenoid valves. The results of the system wireless communication tests indicate that the LoRa-based sensor network has stable data transmission with a maximum communication distance of up to 4 km. At lower communication rates, the signal-to-noise ratio (SNR) and received signal strength indication (RSSI) values measured at long distances are relatively higher, indicating better communication signal quality, but they take longer to transmit. It takes 6 s to transmit 100 bytes at the lowest rate of 0.268 kbps to a distance of 4 km, whereas, at 10.937 kbps, it only takes 0.9 s. The results of field irrigation trials during the wheat grain filling stage have demonstrated that the irrigation amount determined based on the irrigation algorithm can maintain the soil moisture content after irrigation within the suitable range for wheat growth and above 90% of the upper limit of the suitable range, thereby achieving a satisfactory irrigation effect. Notably, the water content in the 40 cm soil layer has the strongest correlation with changes in crop evapotranspiration, and the highest temperature is the most critical factor influencing the water requirements of wheat during the grain-filling period in the test area.

Keywords: EC-IOT; sensor networks; irrigation model; water saving



Academic Editor: Jayme Garcia Arnal Barbedo

Received: 19 December 2024

Revised: 20 January 2025

Accepted: 29 January 2025

Published: 30 January 2025

Citation: Zhang, Y.; Wang, X.; Jin, L.; Ni, J.; Zhu, Y.; Cao, W.; Jiang, X. Research and Development of an IoT Smart Irrigation System for Farmland Based on LoRa and Edge Computing. *Agronomy* **2025**, *15*, 366. <https://doi.org/10.3390/agronomy15020366>

Copyright: © 2025 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

1. Introduction

Population growth, water scarcity, and food shortage are major challenges to global sustainable development. The current world population is around 8 billion, with about 735 million people in a state of chronic hunger. Therefore, it is crucial to accelerate the doubling of agricultural productivity and ensure food security [1,2]. In order to achieve the goal of “Zero Hunger”, the Food and Agriculture Organization of the United Nations (FAO) has listed irrigation technology as one of the six key technologies to enhance food security resilience. Currently, 72% of the world’s freshwater resources are used for agricultural production, with crop irrigation consuming the most, and 41% of the world’s water used for irrigation is used at the expense of the demand for environmental flows [3,4]. For China, the situation is even more serious, irrigation water consumption accounts for 86% of agricultural water consumption, while the effective use coefficient of farmland irrigation water is only 0.576. Moreover, agricultural production is dominated by small-scale farmers, and irrigation management mainly relies on the subjective experience of agricultural producers, lacking precise control methods, which can easily lead to irrational irrigation. Therefore, applying agricultural 4.0 technologies such as the Internet of Things, artificial intelligence, and edge computing to crop irrigation revolutionizes the management of agricultural water use. It enables remote monitoring, automatic control, and precise irrigation, and it manages irrigation water resources effectively and sustainably by providing the exact volume of water needed by crops. These technologies have become a key solution for promoting the intelligence of agricultural irrigation and optimizing water-saving management [5–7].

The “human-machine-thing” interconnection of agricultural IoT aims to achieve real-time data collection, positioning, tracking, and management of agricultural production objects and processes, as well as to assist farmers in more accurately and dynamically managing and controlling different agricultural components, links, and systems. Therefore, the IoT architecture can generally be divided into three layers: the perception layer, the transmission layer, and the application layer [8,9]. For irrigation scenarios, it is necessary to monitor various information indicators such as soil, atmospheric environment, and water flow, and since the area covered is quite large, establishing a reliable data transmission connection at the transmission layer is essential. Most studies mainly address these issues by constructing wireless sensor networks using existing wireless communication technologies such as Wi-Fi, Zigbee, LoRa, and NB-IoT [10]. Goap et al. [11] designed a smart irrigation system based on IoT architecture that integrated Zigbee and Wi-Fi. The system utilized Zigbee to transmit physical parameters from the agricultural fields and established a connection with a web server using Wi-Fi, effectively covering the irrigation area network. Ramson et al. [12] developed a soil health monitoring system whose self-powered terminal nodes utilized LoRaWAN (wide area network) to achieve long-distance transmission of information. To achieve the management of a smart irrigation system for vineyards. Liopa-Tsakalidi et al. [13] used NB-IoT to transmit data collected by microcontrollers to the application layer so that the irrigation execution module could adjust and control based on the data. As mentioned earlier, various wireless communication technologies are available for information transfer and Table 1 compares different communication technologies. For large-scale farmland irrigation, LoRa is the optimal solution. It has been widely used in monitoring mountain orchards, fishery water quality, and pasture farming. It is suitable for large-scale deployment due to its simple networking, long communication distance, high interference immunity, and ultra-high reception sensitivity [14–17].

Table 1. Comparison of different communication technologies.

Parameters/Technologies	Bluetooth	Wi-Fi	ZigBee	LoRa	NB-IOT
IEEE standard	802.15.1	IEEE802.11	802.15.4	802.15.4 g	3GPP release13
Data rate	1 Mbps	11–54 Mbps	20–250 kbps	50 kbps	160–200 kbps (UL) 160–250 kbps (DL)
Frequency band	2.4 GHz	2.4 GHz	868/915 MHz and 2.4 GHz	869/915 MHz	LTE frequency
Nominal range	20–100 m	100 m	100 m	Urban: 2–5 km suburban: 15 km	Urban: 1 km suburban: 10 km
Advantage	Low price and simple configuration	High flexibility	Low power consumption and flexible network topology	Strong anti-interference and stable operation	Large-scale deployment and data security
Disadvantage	limited coverage area	High power consumption	Low interference resistance	Low data transfer rate	High deployment cost

In addition to ensuring the reliable data transmission from the irrigation site, modern smart irrigation systems must also integrate advanced irrigation algorithms to achieve precise irrigation. At present, in the field of water-saving irrigation, various control algorithms such as machine learning, fuzzy control, and evapotranspiration (ET) have been introduced. Benzaouia et al. [18] proposed a feedback fuzzy logic control method combined with soil and weather conditions to determine the irrigation timing for crops, keeping soil moisture above a predetermined threshold. Peng et al. [19] constructed the acquired field environmental information to construct a crop water requirement prediction model based on the back propagation (BP) neural network. Vianny et al. [20] proposed a hybrid irrigation forecasting model based on machine learning algorithms such as K-nearest neighbors (KNN), gradient boosting tree (GBT), and long short-term memory (LSTM), which could predict the water requirements of crops at different time intervals. However, applying machine learning to irrigation systems has the drawback that constructing predictive models requires high-quality data sources to ensure effective learning rates and prediction accuracy. Collecting datasets that affect crop quality and water requirements throughout the crop irrigation cycle is challenging, and models may become ineffective when crops, soils, environments, and irrigation conditions differ [21]. Another widely used method for water saving is to calculate the volume of irrigation based on the ET values, which refer to evaporation from the soil surface and the transpiration from the plant canopy and are affected by a variety of factors such as meteorology, crop growth stage, and plant density [22]. Compared to time-based irrigation schedules, ET-based irrigation systems can save up to 42% of water use, according to Davis et al. [23]. Dela et al. [24] designed an automatic irrigation system that utilized the Penman–Monteith method to calculate the evapotranspiration of a shepherd’s purse, saving 71.81% of water compared to manual irrigation systems. Some studies have also proposed combining the PM equation with fuzzy logic to design an irrigation system, using the evapotranspiration values calculated by the PM equation and converting them to irrigation volumes through fuzzy logic as a way to improve water use efficiency [25]. The downside is that using ET to design irrigation systems requires more meteorological data and specialized agronomic knowledge.

Currently, the processing and storage of algorithms and data are mainly concentrated in the cloud. If the node data volume is large and far away, it will cause significant bandwidth pressure, increase latency, and is not conducive to data privacy protection [26,27]. Edge computing, a distributed processing model that performs compu-

tations closer to the data source, has emerged as a new solution [28]. Schneider Electric analyzed the cost-effectiveness of edge computing deployment and found that edge computing data centers saved 42% on investment costs compared to cloud computing data centers [29]. Moreover, studies have shown that branched neural network models based on edge computing reduce inference latency by an average of 36% compared to cloud deployments [30]. Nowadays, edge computing applications in agriculture mainly focus on environmental monitoring and pest identification, crop growth and yield prediction, and agricultural scouting and path planning, while less research has been performed for irrigation [31,32].

In summary, this study proposed an ET-based smart irrigation system for farmland using LoRa and EC-IoT architecture, achieving the design of hardware and software for terminal nodes and multi-mode gateways and establishing a star topology sensor network. The gateway integrated an irrigation algorithm as an edge computing terminal. It used real-time meteorological sensor data on-site to calculate the required irrigation volume for crops using the Penman–Monteith equation and the soil water balance equation. It also used the lower limit of the soil moisture content set according to the soil water retention characteristics and the crop growth stage as the trigger condition for irrigation scheduling. A real-time visualization application was developed for monitoring irrigation and field data that could remotely issue commands to control the opening and closing of the pulse valve and irrigate according to the user-defined irrigation volume. Some of the innovations of this research are as follows: (1) the system proposes a new architecture for a smart irrigation system in farmland based on LoRa and edge computing gateways, which reduces latency and cloud deployment server costs. (2) the irrigation system considers soil characteristics, crop growth stages, and meteorological information to calculate the irrigation volume and can change the irrigation schedule based on real-time meteorological information.

2. Materials and Methods

Figure 1 illustrates the overall structure of the irrigation system, composed of three parts: nodes, edge computing gateways, and the application end. The node part primarily stores the acquired sensor information, sends it to the gateway, and controls the pulse solenoid valves. The gateway is the core of the system, equipped with edge computing capabilities. It is responsible for uploading node information and the collected meteorological data to the cloud and executing irrigation algorithms for scheduling irrigation. The application end provides for users with an interface for visualizing field data, allowing for data storage and remote control of the pulse valves.

The system uses sensors such as soil moisture sensors (Shandong Renke Control Technology Co., Ltd., Jinan, China), magnetic flow meters (Kaifeng Instrument Co., Ltd., Kaifeng, China), air temperature and humidity sensors, sunshine duration sensors, wind speed sensors, rain gauges, and atmospheric pressure sensors (Hebei Ousu Electronic Technology Co., Ltd., Shijiazhuang, China). Table 2 provides the technical parameters for each sensor. All the sensors used are manufactured by sensor companies in China. Their working principles are as follows: Soil moisture sensors are based on time-domain reflectometry (TDR), which calculates the moisture content of the soil by measuring the propagation time of electromagnetic waves in the soil. Magnetic flow meters apply the principle of electromagnetic induction to measure the conductive liquid flow by detecting the electromotive force (EMF) generated when the liquid passes through an applied magnetic field. Air temperature and humidity sensors are based on the capacitive principle. When the temperature and humidity in the air change, the capacitance value also changes. Therefore, the temperature and humidity values in the air can be calculated by measuring the capacitance value. The working principles of the sunshine duration sensors

are based on high-precision photosensitive elements and preset threshold voltage signals. When sunlight shines on the photosensitive elements, they generate voltage signals. When the radiation voltage value on the instrument exceeds the preset threshold (200 mV), the sensors will accumulate the time. Wind speed sensors are propeller-type, which calculate the wind speed by measuring the driving force of the wind on the propeller. Rain gauges are based on the principle of the tipping bucket and electronic signal conversion technology. When rainwater falls on the tipping bucket, it turns over at a certain angle, which triggers the sensor to generate an electrical signal, and the counter calculates the total amount of precipitation by accumulating the number of tipping bucket tips. Atmospheric pressure sensors utilize the characteristic of the pressure-sensitive elements within the sensor, where the resistance value changes with pressure, to measure atmospheric pressure.

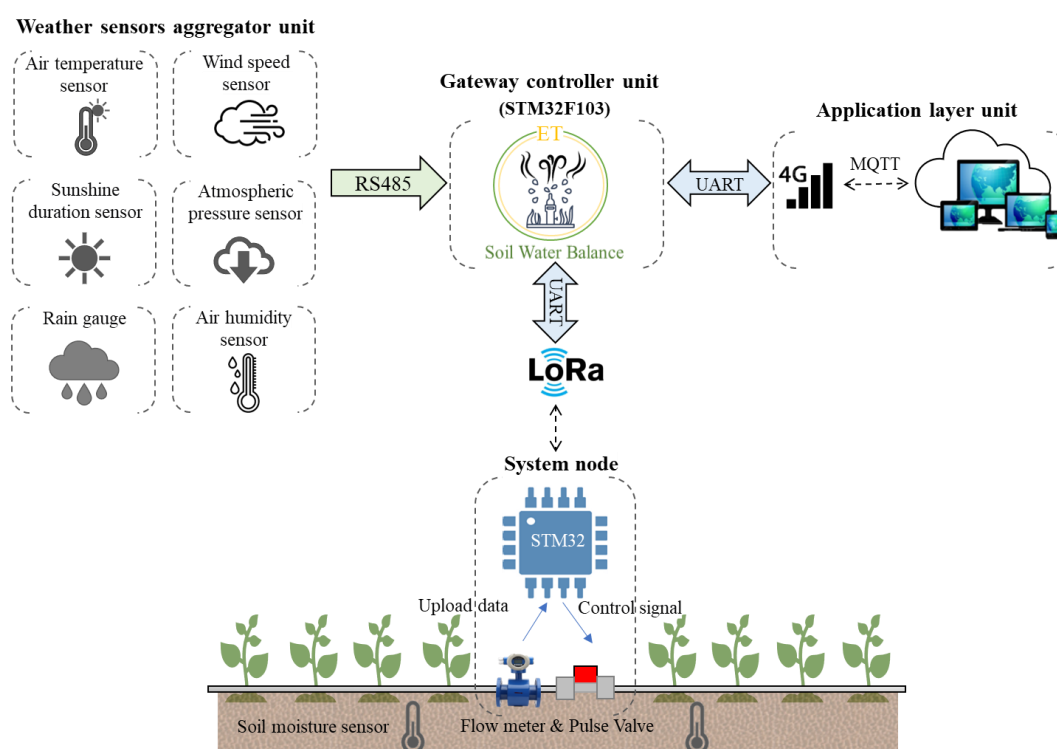


Figure 1. Schematic of overall irrigation system. In the figure, UART stands for universal asynchronous receiver/transmitter, STM32 stands for the STM32F103ZET6 (Milpitas, CA, USA) microcontroller, RS485 is a serial communication standard used for data transmission between devices over long distances.

Table 2. Technical parameters of sensors. RH stands for relative humidity, R stands for relative error.

Sensor Type	Working Voltage	Scale Range	Measuring Accuracy
Soil moisture sensor	DC 5–12 V	0–100% RH 0–50 cm	$\pm 2\%$ within 0–50% $\pm 3\%$ within 50–100%
Magnetic Flow Meter	DC 12 V	2.2–22.0 m ³ /h	$\pm 0.5\%R$
Air temperature sensor	DC 5–24 V	−30–70 °C	± 0.2 °C
Air humidity sensor	DC 5–24 V	0–100% RH	$\pm 3\%$ RH
Sunshine duration sensor	DC 5–24 V	0–24 h	$\pm 2\%$ h
Wind speed sensor	DC 5–24 V	0–50 m/s	$\pm (0.3 + 0.03 V)$ m/s

Table 2. Cont.

Sensor Type	Working Voltage	Scale Range	Measuring Accuracy
Rain gauge	DC 5–24 V	0–100 mm/day	$\pm 3\%$
Atmospheric pressure sensor	DC 5–24 V	10–1100 hPa	± 0.5 hPa

2.1. Hardware Design

2.1.1. Node Hardware Design

The nodes of the smart irrigation system integrate sensing and control and are deployed in the farmland away from the gateway. The nodes need to be capable of collecting soil moisture information, irrigation flow, and location data, automatically controlling pulse valves, and storing and transmitting data. Therefore, the hardware design of the nodes mainly consists of the master control module, data acquisition module, LoRa communication module, storage module, and valve control module. The block diagram of the nodes' hardware system design is shown in Figure 2, and the hardware circuit diagram is shown in Figure 3.

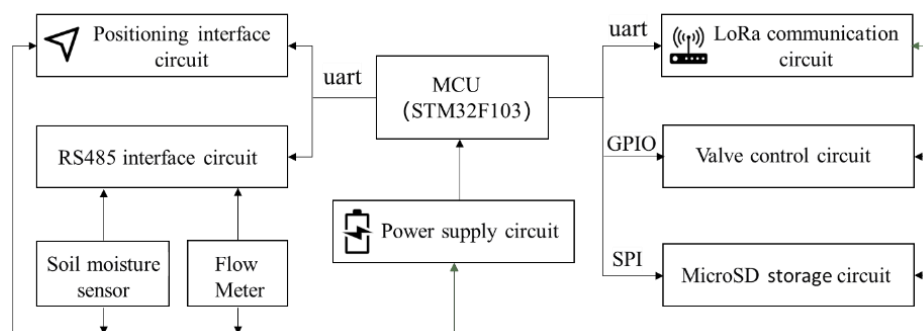


Figure 2. Schematic diagram of node hardware system design. General purpose input/output (GPIO) is a general purpose input/output port, a general purpose pin on a microcontroller; serial peripheral interface (SPI) is a high-speed, full-duplex, synchronous serial communication protocol used for communication between microcontrollers and peripheral devices.

STM32 master control module: The master control chip for the node is the STM32F103ZET6 (Milpitas, CA, USA), a 32-bit MCU based on the ARM Cortex-M3 core with a clock frequency of 72 MHz. It has a rich set of communication interfaces such as universal synchronous/asynchronous receiver/transmitter (USART), SPI, and inter-integrated circuit (I2C), supports 8 timers, and has multiple interrupt source configurations, meeting the needs for data interaction and processing between various modules of the node [33].

Data acquisition module: The integrated 485 isolation chip SIT3485ISO (Hunan Silicon IoT Technology Co., Ltd., Changsha, China) is used to design 4-channel RS485 interfaces connected to the microcontroller's USART2 to achieve multi-channel acquisition for soil moisture sensors and flow meters. An NMOS transistor (STMicroelectronics, Geneva, Switzerland) is also employed to add a switch power supply in order to reduce power consumption.

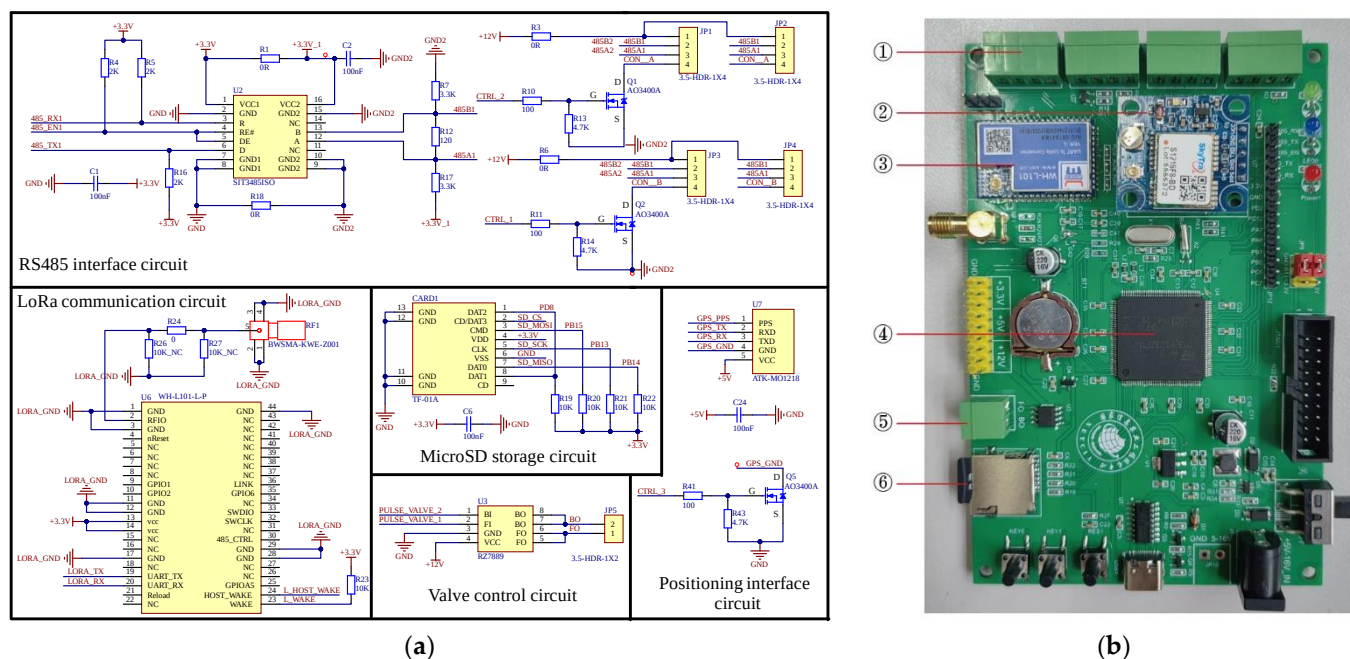


Figure 3. Node hardware circuit diagram. (a) Circuit schematic diagram. (b) PCB board: ① RS485 data acquisition interfaces; ② positioning module; ③ LoRa communication module; ④ STM32 master control module; ⑤ valve control interface; ⑥ storage module.

LoRa communication module: The system uses a LoRa serial communication module (WH-L101-C, Uxr, Jinan, China) designed based on the Sx1268 chip from Semtech Corporation (Camarillo, CA, USA), operating in the 410–525 Mhz band. The LoRa module communicates with the MCU via USART3, and an on board SMA RF (Zhenjiang Dongming Telecom Technology Co., Ltd., Zhenjiang, China) antenna interface is designed for it. The main function is to upload the soil moisture content, node location information, irrigation flow, and valve status information processed in real-time by the MCU to the gateway in accordance with the specified frame format and to receive the commands issued by the gateway to collect sensor information and control valve operations.

Storage module: The node stores information on a MicroSD card (SanDisk, Milpitas, CA, USA), which facilitates data sharing among different devices. The system's master control chip writes sensor data and irrigation data to the SD card via an SPI channel and has ported the FatFs file system for data management, addressing the issue of data loss during LoRa communication that could result in the inability to record valid data.

Valve control module: The irrigation valves utilize low-power pulse solenoid valves. Since the GPIO pins of the master control chip only provide a current of ± 8 mA, which is insufficient to directly drive the solenoid valve, a RZ7889 IC chip (Wuxi Smart Microelectronics Co., Ltd., Wuxi, China) has been employed to enhance the driving capability and output bi-directional 12 V, 50 ms pulse signals to control the opening and closing of the valve. To prevent communication errors between the node and the gateway that could render the valve inoperable for closing or opening, manual control via buttons has also been incorporated.

2.1.2. Gateway Hardware Design

The edge computing gateway of the smart irrigation system serves as the central component, connecting LoRa nodes with the cloud platform, collecting meteorological data, analyzing and processing information, and running irrigation algorithms for scheduling irrigation to reduce cloud deployment costs. Consequently, the system's gateway must support a variety of communication protocols and data formats and possess high-performance

computing capabilities. The hardware system design diagram of the gateway designed in this study is shown in Figure 4, and the hardware circuit diagram is shown in Figure 5. The gateway also selects the STM32F103ZET6 chip as the master control core and designs four communication circuits to meet various communication needs. Since the meteorological station and the gateway are integrated, the gateway also uses the SIT3485ISO chip to design RS485 data acquisition interfaces in the hardware circuit for connecting to meteorological sensors. The following section focuses on the LoRa communication module, 4G communication module, Ethernet communication module, storage module, and power module of the gateway.

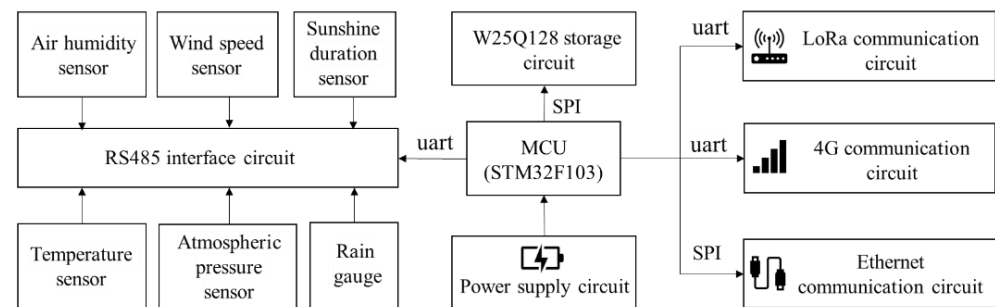


Figure 4. Schematic diagram of gateway hardware system design.

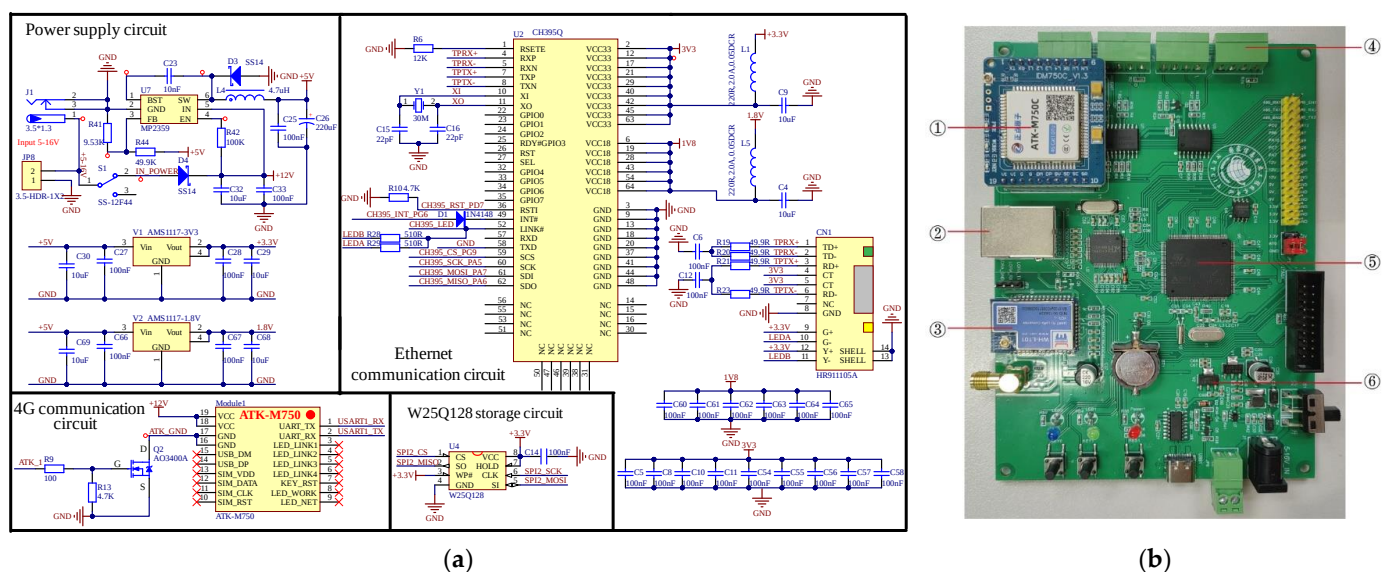


Figure 5. Gateway hardware circuit diagram. (a) Circuit schematic diagram. (b) PCB board: ① 4G communication module; ② Ethernet communication module; ③ LoRa communication module; ④ RS485 data acquisition interfaces; ⑤ STM32 master control module; ⑥ power module.

LoRa communication module: The gateway uses the same type of LoRa module as the node, and the main controller uses the USART3 peripheral to configure them to the same operating frequency band and mode using attention (AT) commands. In order to coordinate simultaneous communication between multiple nodes and the gateway, the gateway system uses a time-slicing polling method to access the nodes in the network. It sends a command to the nodes to collect data. If this irrigation system is determined by the schedules that the node needs irrigation, the specific node will be actively issued to the irrigation command.

4G communication module: Due to the data stream nature of the information collected by the field sensors, 4G communication was selected for its cost-effectiveness, utilizing the ATK-IDM750C module (Alientek, Guangzhou, China). This module supports various

application-layer communication protocols and is compatible with 3G and 2G networks. The master controller communicates with the 4G module via the USART1 peripheral using AT commands and utilizes the MQTT protocol for information exchange with the cloud server [34].

Ethernet communication module: The gateway system has expanded Ethernet capabilities by incorporating the CH579Q network interface chip (Nanjing Qinheg Microelectronics Co., Ltd., Nanjing, China). This chip integrates the Ethernet media access control (MAC) layer and network protocol stack and interacts with the master controller via the SPI1 channel. Using a category 6 (CAT-6) cable with a registered Jack-45 (RJ-45) interface, the gateway can be directly connected to the Internet.

Storage module: The gateway master control chip has limited flash memory capacity. Hence the W25Q256 chip is utilized as an external storage expansion for storing real-time generated node communication data and programs, among others. The master controller performs data read and write operations on it via the SPI2 channel.

Power module: Due to insufficient power supply facilities in the farmland, both the gateway and the nodes are powered by a solar power generation system. The power supply battery uses a 12 V 18650 lithium battery pack with a capacity of 18,000 mAh. Since the positioning module and serial debugging module on the circuit board require 5 V power supply, the microcontroller and LoRa module require 3.3 V power supply, and some electronic devices such as light-emitting diodes require 1.8 V power supply, it is necessary to design voltage regulation circuits. In this paper, we use voltage regulator chips of models MP2359 (Monolithic Power Systems Inc., Kirkland, WA, USA), AMS1117-3.3V, and AMS1117-1.8V (AMS AG, Vienna, Austria) to regulate the 12 V voltage to 5 V, 3.3 V, and 1.8 V, respectively, to provide power to the circuits of various system modules.

2.2. Software Design

2.2.1. Node Software Design

The software program of the node mainly consists of five parts: data acquisition program, storage program, valve control program, data packet reception and transmission program, and power consumption management program. The flowchart of the node software design is shown in Figure 6. Initially, upon powering up, each module undergoes initialization, after which the positioning module is analyzed to determine the location of the node. The CPU then transitions into the STM32's low-power sleep mode. It is awakened under two specific conditions: the first is when the LoRa module receives data and USART generates a receive idle interrupt; the second is when the timer triggers an overflow interrupt during the intermittent reading of the flow meter data after irrigation is started.

The data acquisition program initiates upon receiving the polling frame from the gateway via the serial port, employing the Modbus-RTU protocol to interrogate the moisture content across various soil layers of soil sensors, each identified by unique slave addresses. Once parsed, the data are transmitted to the gateway as a string, adhering to a frame structure encompassing a header, footer, node address, soil moisture data, and geolocation data. The storage program leverages the FatFs file management system to generate CSV files on the SD card, archiving details such as date, time, moisture content across different soil strata, geolocation coordinates, and irrigated water volume for subsequent review on other devices. The valve control program is designed with two modes: manual control and automatic control. Under automatic control, the program initiates irrigation after parsing the irrigation command transmitted by the gateway. Utilizing the overflow interrupt flag of a defined timer, the program reads the flow meter's cumulative flow every 30 s and compares it against the designated irrigation volume. Once the target irrigation volume

is achieved, the valve and timer are deactivated, and subsequently, data are uploaded, including post-irrigation soil moisture levels and the volume of water irrigated at the node.

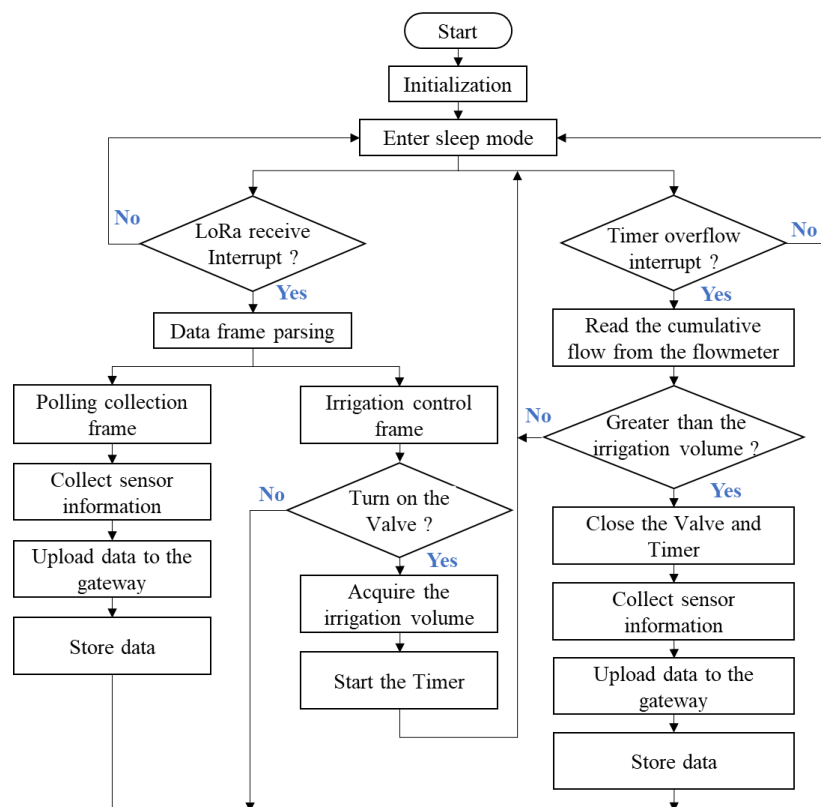


Figure 6. Node software design flowchart.

In summary, the nodes are primarily responsible for collecting soil moisture information and field location information; after processing and local storage, the data are reported to the higher-level gateway. They can also receive commands from the gateway to control the opening and closing of water valves, execute irrigation operations, and after the irrigation is completed, upload the soil moisture content and irrigation volume after irrigation to the gateway.

2.2.2. Gateway Software Design

The edge computing gateway belongs to the middle layer in the system framework, offering local services to multiple nodes, reducing the latency associated with remote operations from the MQTT server and alleviating the bandwidth load. Its software design, based on the FreeRTOS operating system and the Keil MDK5 software development platform, integrates data transmission and irrigation scheduling into one system. The FreeRTOS operating system, based on priority-based preemptive scheduling, provides harmony multi-tasking capacity, thereby improving the real-time performance of the gateway system's data transmission and irrigation scheduling. The workflow diagram of the gateway system is shown in Figure 7. The gateway system commences operation by initially configuring the peripherals and system clock of the STM32 microcontroller. Subsequently, it uses xTaskCreate to create the polling node task, the meteorological data acquisition task, the task for the LoRa module to receive and send data, the task for the 4G module to receive and send data, and the task related to irrigation scheduling while assigning priorities to these newly created tasks. Multiple tasks communicate through task priorities, event flag groups, and queues for task switching, synchronization, and message

passing. This orderly nesting prevents the program from running idle, thereby greatly improving the program's execution efficiency.

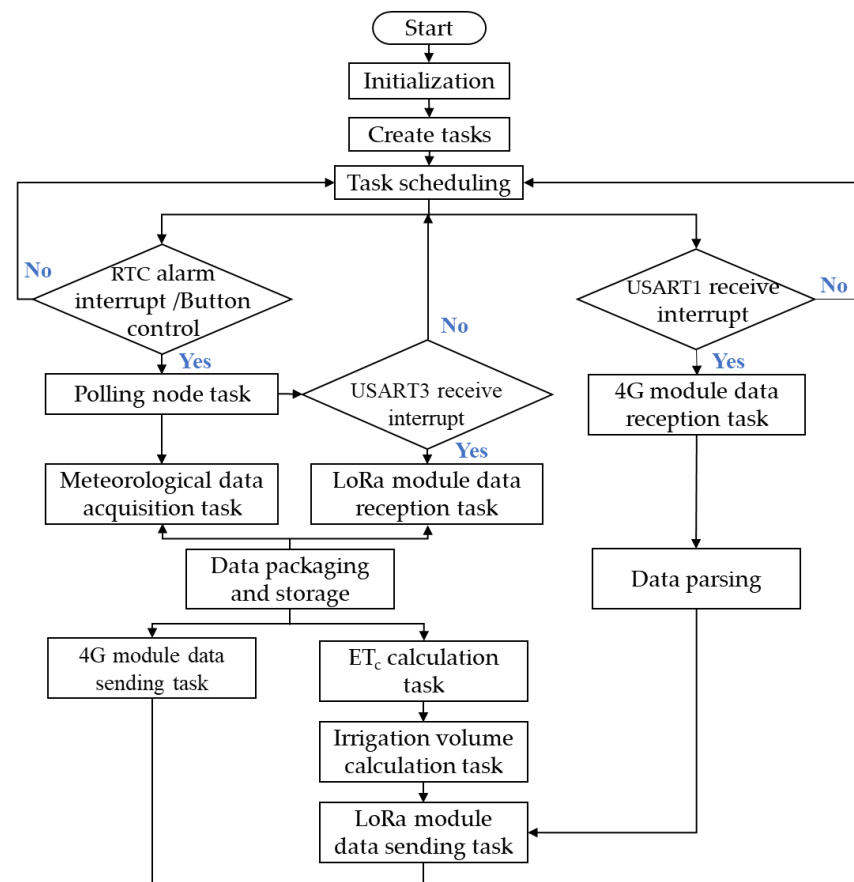


Figure 7. Gateway software design flowchart. Real-time clock (RTC) refers to a real-time clock, which is an electronic device used for maintaining time and date; ET_c refers to the actual evapotranspiration of crops.

The 4G module reception task involves parsing instructions from the cloud for controlling valve switches or irrigation volumes, and it has the highest task priority. This task waits for an idle interrupt to be triggered on USART1, writing the received cloud information into the 4G module reception queue. Once the task reads information from the first in, first out (FIFO) data buffer in the queue, it transitions from a blocked state to a running state. It then parses the instructions by the cloud platform sent using the cJSON library, identifies the corresponding node address with the unique device identifier, and writes the data frames sent to the node into the LoRa transmission queue, thereby triggering the LoRa transmission task.

The polling node task regularly sends commands to various nodes, polling soil moisture and location information of nodes in the self-organizing network. The gateway system can set the polling task event flag through two methods: button or alarm interrupt. The task waits with the event flag bit set to 1, and upon successful waiting, the task transitions from a waiting state to a running state. The feedback from the nodes is written into the LoRa module receive queue through the USART3 idle interrupt, triggering the LoRa module reception task to parse the data. After completing the node polling task, it synchronizes with the meteorological information acquisition task by setting an event flag. At the end of the collection, the meteorological information and node data are uploaded to the cloud and written into the W25Q256 storage. The irrigation scheduling tasks are triggered based on a set irrigation cycle, waiting for event flags to initiate operation. These tasks read node

information and meteorological data stored in the W25Q256 memory to calculate ET_c and the irrigation volume for each node and then write the irrigation commands into the LoRa transmission queue to be sent to the corresponding nodes in real-time.

In summary, the gateway is the core of the system, connecting the downstream nodes to the upstream cloud server. It is primarily responsible for regularly sending information requests to each node, collecting various data from the meteorological sensors, and processing and storing the data returned by the nodes along with meteorological information before uploading it to the cloud. Based on the soil moisture content returned by the nodes, the gateway system determines whether irrigation is needed. If the soil moisture content is lower than the preset threshold, the gateway triggers the irrigation operation, calculates the irrigation amount through the irrigation model, and sends it to the corresponding field. When a user wants to remotely irrigate crops, he/she can click the irrigation button on the visualization page. After receiving the irrigation command, the gateway parses the command, identifies the field that the user wants to irrigate, and sends the command to the corresponding node to complete the irrigation operation.

2.3. ET-Based Decision Algorithm for Irrigation

Establishing a smart irrigation decision algorithm is key to calculating the volume of irrigation. This study proposes a mathematical decision algorithm based on the modified Penman–Monteith (PM) formula and the soil water balance equation, focusing on winter wheat. The workflow of the algorithm is shown in Figure 8. The initial irrigation is based on the modified PM formula calculated according to real-time meteorological information to obtain that day's reference evapotranspiration (ET_0). The upper and lower limits of suitable soil moisture content for each growth stage are determined by referring to the water requirements of wheat and the soil bulk density. If the real-time measured soil moisture content is below the lower limit, irrigation is required. First, the water requirement of the wheat is calculated, and then the irrigation volume is derived using the soil water balance equation. The subsequent irrigation is a form of supplementary watering. Irrigation is initiated if the real-time soil moisture content measured during the monitoring period falls below the lower limit. The difference between the accumulated crop water requirements and the accumulated precipitation determines the volume of irrigation.

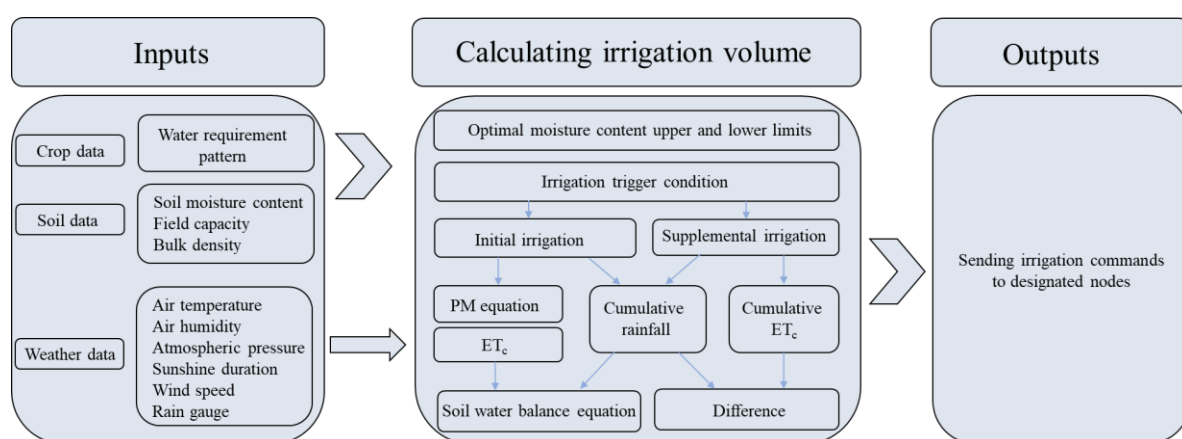


Figure 8. Workflow diagram of the ET-based irrigation decision algorithm.

The water requirements of winter wheat are influenced by its growth conditions and environmental factors. During its growth period, there are four key periods of water requirement: overwintering water, regreening water, jointing, and filling water, which have a significant impact on wheat yield and quality [35]. Table 3 shows the suitable soil moisture conditions for winter wheat growth during each growth stage [36]. From the

emergence of wheat seedlings to the tillering stage, the soil moisture should not be lower than 70% of the field capacity. Otherwise, it will affect the seedling emergence rate, delay the emergence, and reduce the number of tillers before winter. From the overwintering stage to the regreening stage, the soil moisture generally cannot be lower than 70% from when the above-ground stems and leaves of the wheat die to when new leaves grow back in the spring. Sufficient moisture is extremely beneficial for inhibiting soil evaporation and protecting the tillers from wintering safely, and ensuring the suitable moisture for regreening can strive for more spikes and grains, which is conducive to high yield. From the jointing to the mid-grain filling stage, the growth rate of wheat accelerates. Higher soil moisture conditions benefit photosynthesis, which is used for the growth of stems and spikes, as well as grain filling. The soil moisture should not exceed 80% of the field capacity during this stage. From the late grain filling to the maturity stage, dry matter accumulation slows down in the late grain filling stage, and the grain body begins to contract. Hence, soil moisture at 55–60% of the field capacity is preferable. Excessive moisture conditions are not conducive to grain maturation and may even lead to late maturity, decreasing thousand-kernel weight. Generally, no irrigation is needed during the maturity stage.

Table 3. Suitable soil relative volumetric moisture content for each growth stage of winter wheat.

Wheat Growth Stages	Emergence– Tillering	Overwintering– Regreening	Jointing– Mid-Grain Filling	Late Grain Filling
Upper limit of field capacity (%)	80	85	85	60
Lower limit of field capacity (%)	70	70	80	55

Currently, there are many methods for estimating ET_0 , such as the Hargreaves equation, PM, and Turc-Radiation, etc. [37]. The PM formula is the standardized method the Food and Agriculture Organization (FAO) recommends for calculating ET_0 , involving meteorological data such as solar radiation, temperature, humidity, wind speed, and atmospheric pressure [22]. The calculation formula is as follows:

$$ET_0 = \frac{0.408\Delta(R_n - G) + \gamma \cdot \frac{900}{(273+T)} \cdot u_2(e_s - e_a)}{\Delta + \gamma(1 + 0.34u_2)} \quad (1)$$

Among them,

Δ —Slope vapor pressure curve (kPa/°C);
 R_n —Net radiation at the crop surface (MJ/(m²·day));
 G —Soil heat flux density (MJ/(m²·day));
 γ —Psychrometric constant (kPa/°C);
 T —Mean daily air temperature at 2 m height (°C);
 u_2 —Wind speed at 2 m height (m/s);
 e_s —Saturation vapor pressure (kPa);
 e_a —Actual vapor pressure (kPa).

Where T , u_2 are directly measured by sensors, and all other parameters, except for R_n , are calculated using the formulas recommended by FAO-56.

$$R_n = R_{ns} - R_{nl} \quad (2)$$

$$R_{ns} = (1 - 0.23)R_s \quad (3)$$

Among them,

R_{ns} —Net solar radiation (MJ·m^{−2}·d^{−1});

R_{nl} —Net longwave radiation ($\text{MJ}\cdot\text{m}^{-2}\cdot\text{d}^{-1}$);
 R_s —Solar radiation ($\text{MJ}\cdot\text{m}^{-2}\cdot\text{d}^{-1}$).

To enhance the accuracy of crop ET_0 estimation and make the PM formula more applicable to the application area, we improved the calculation of R_s by referring to the simulation method of daily solar radiation proposed by Tong et al. [38]. This method uses conventional meteorological monitoring data to simulate solar radiation, with the monitoring parameters including latitude, longitude, altitude, and sunshine hours. First, it is necessary to calculate the extraterrestrial radiation H_0 , and the calculation formula is as follows:

$$H_0 = \frac{G_{sc} \cdot E_0 (\cos \phi \cdot \cos \delta \cdot \sin W_s + \frac{\pi}{180} \cdot \sin \phi \cdot \sin \delta \cdot W_s)}{\pi} \quad (4)$$

where G_{sc} is the solar constant, whose value is $118.108 \text{ MJ}\cdot\text{m}^{-2}\cdot\text{d}^{-1}$; E_0 is the Earth orbital eccentricity correction factor; ϕ is the latitude; δ is the solar declination; and W_s is the sunset hour angle. Their calculation formula is as follows:

$$E_0 = 1.00011 + 0.034221 \cos \tau + 0.00128 \sin \tau + 0.000719 \cos 2\tau + 0.000077 \sin 2\tau \quad (5)$$

$$\delta = \frac{180}{\pi} \cdot (0.006918 - 0.399912 \cos \tau + 0.070257 \sin \tau - 0.006758 \cos 2\tau + 0.000907 \sin 2\tau - 0.002697 \cos 3\tau + 0.00148 \sin 3\tau) \quad (6)$$

$$\tau = \frac{2\pi \cdot (n - 1)}{365} \quad (7)$$

$$W_s = \cos^{-1}(-\tan \phi \cdot \tan \delta) \quad (8)$$

$$H_L = 0.8 \cdot H_0 \quad (9)$$

In Formula (7), τ represents the annual angle, and n is the day sequence number. H_0 is partially absorbed by the atmosphere as it passes through to reach the ground. Under clear-sky conditions, the total radiation in the atmosphere has a transparency coefficient of about 0.8. Thus the total ground radiation (H_L) under clear-sky conditions can be derived, as shown in Formula (9). Finally, the Angstrom–Prescott equation can be used to calculate solar radiation R_s . The calculation formula is as follows [38]:

$$R_s = H_L \cdot \left(a + b \cdot \frac{S}{S_L} \right) \quad (10)$$

$$S_L = \frac{2}{15} \cdot W_s \quad (11)$$

where S represents the sunshine duration measured by the sensor; S_L is the daylight hours; a and b are empirical parameters obtained from a regression simulation based on the measured solar radiation, a is 0.248 and b is 0.752, cited from Zuo et al. [39] based on the monthly average of the measured total radiation, the percentage of sunshine, and the monthly total radiation under the condition of a clear sky in different types of areas in China.

Due to the differences in ground cover, canopy characteristics, and aerodynamic resistance of wheat compared to the reference crop used for calculating reference evapotranspiration (ET_0), the actual evapotranspiration (ET_c) of wheat needs to be modified using a crop coefficient (K_c). This study employs the single crop coefficient method, as shown in Formula (12), where K_c is the crop coefficient recommended by FAO-56 [22] for different growth stages of winter wheat, as shown in Table 4.

$$ET_c = K_c \cdot ET_0 \quad (12)$$

Table 4. Crop coefficient.

Growth Stages	Initial Stage	Middle Stage	End Stage
K_c	0.7	1.15	0.4

After obtaining ET_c based on the modified PM formula, wheat’s required initial irrigation volume during the measured period can be calculated using the soil water balance equation [40]. Without considering the replenishment from groundwater, the water balance equation is as follows:

$$W_t - W_0 = W_r + P_0 + M - ET_c \quad (13)$$

Among them,

- W_0 —The soil water storage of the planned wetted layer at the initial moment (mm);
- W_t —The soil water storage of the planned wetted layer at the end of the moment (mm);
- W_r —The additional soil water storage due to the increase in the root zone (mm);
- ET_c —The evapotranspiration of wheat during the period (mm);
- M —Irrigation volume (mm);
- P_0 —Effective rainfall (mm).

After the initial irrigation, due to root absorption and evapotranspiration, the water storage in the soil wetted layer gradually decreases. When it falls below the lower limit of the suitable moisture content, supplementary irrigation is triggered. The calculation formula for the irrigation volume is as follows.

$$M = \sum_{i=1}^n ET_{ci} - \sum_{i=1}^n P_i \quad (14)$$

In Formula (14), ET_{ci} is the crop evapotranspiration on day i ; P_i represents the rainfall on day i ; and n is the number of days since the last irrigation. Therefore, the irrigation volume is determined by the difference between the cumulative crop evapotranspiration and the cumulative rainfall over these (n) days. It is important to note that the formula for supplemental irrigation applies during the same growth stage of wheat when the root zone is relatively constant. When wheat enters a new growth stage and the root zone increases, the irrigation amount needs to be calculated according to Formula (13).

2.4. Visualization

The system visualization interface is provided by WeChat mini programs, which are applications that users can use without the need to download or install. This application is primarily built and developed based on WeChat Developer Tools and JavaScript while utilizing WXML, WXSS, JSON, and the Apache ECharts plugin for auxiliary programming. The mini program has four functional pages, as shown in Figure 9. Among them, Figure 9a displays real-time meteorological information in the field to users. Figure 9b shows the soil moisture content at different soil depths of each node’s field in real-time. Where layer 1–4 refers to the soil depth from 20–50 cm, if the user needs to monitor the soil moisture content at a deeper depth, more components can be added to this page. Figure 9c uses charts to display the trend of soil moisture content changes in each node’s field. Due to page size limitations, only the historical data of the most recent 10 soil moisture uploads can be displayed. Figure 9d is used for remote irrigation control, allowing users to control the opening and closing of valves and send down custom irrigation volume commands by clicking on the button component. Users can click the “Open” or “Close” button in the control box of each node to remotely control the opening and closing of the water valve.

Additionally, users can enter a specific irrigation amount in the input field below and click the “Send” button. The corresponding field will then be irrigated according to the irrigation amount entered by the user.

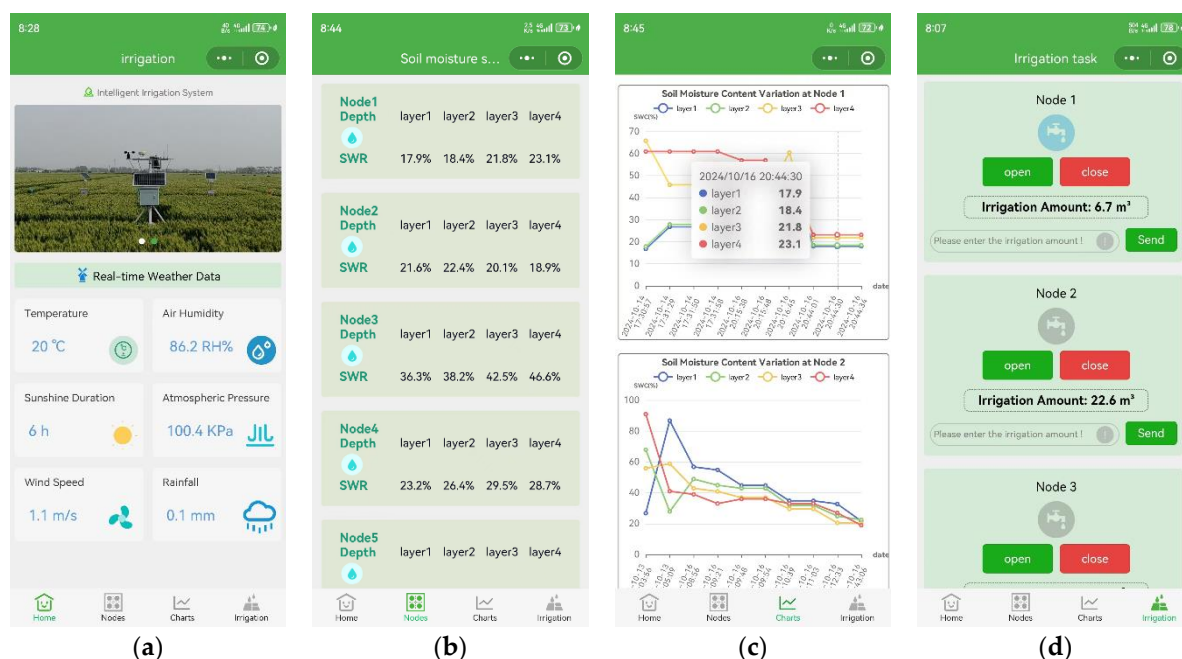


Figure 9. The system visualization interface. (a) Meteorological data display page. (b) Soil moisture content display page. (c) Soil moisture content trend display page. (d) Irrigation task page.

The main workflow of the mini program is as follows: The mini program first establishes a connection to the MQTT cloud server via the WebSocket Secure (WSS) protocol. Once the connection is successful, it constructs the upstream and downstream links by subscribing to and publishing different topics. The MQTT upstream link subscribes to real-time information from different node fields on the cloud server. By listening to messages on the page and analyzing and processing them, the latest data are updated on the mini program page. Users can click the open/close button on the page or enter a custom irrigation amount and click the send button. The MQTT downlink can then send the corresponding control command to the cloud server, which indirectly sends the control command to the system gateway to control the opening and closing of the valve.

3. Demonstration and Tests of the Smart Irrigation System

3.1. System Communication Tests

Field tests for the communication coverage range of system nodes and gateways were conducted at the Baima Teaching and Research Base of Nanjing Agricultural University in Jiangsu Province. The test mainly included the maximum communication distance (without a relay gateway) and communication quality of the system's LoRa module at different transmission rates, providing a reference for users to deploy wireless sensor networks in real-world application scenarios. On-site deployment is as shown in Figure 10: the gateway is placed at a fixed location to operate and send data packets continuously, and the nodes are deployed in the test area at intervals of 500 m away from the gateway. The area includes common elements in both urban and field environments, with trees, crops, buildings, roads and cars. The gateway LoRa module is set to a transmission power of 21.5 dBm, with an antenna gain of 3 dBi, and is mounted at a height of 2 m. At a working frequency of 475 MHz, it sends 100 packets of 100 bytes in size to nodes at various locations at different transmission rates of 0.268 kbps, 0.537 kbps, 0.977 kbps, 3.125 kbps, 10.937 kbps,

and 37.5 kbps. The test results recorded the number of data packets received, the received signal strength indicator (RSSI) values, the signal-to-noise ratio (SNR), and the transmission time at eight test points.

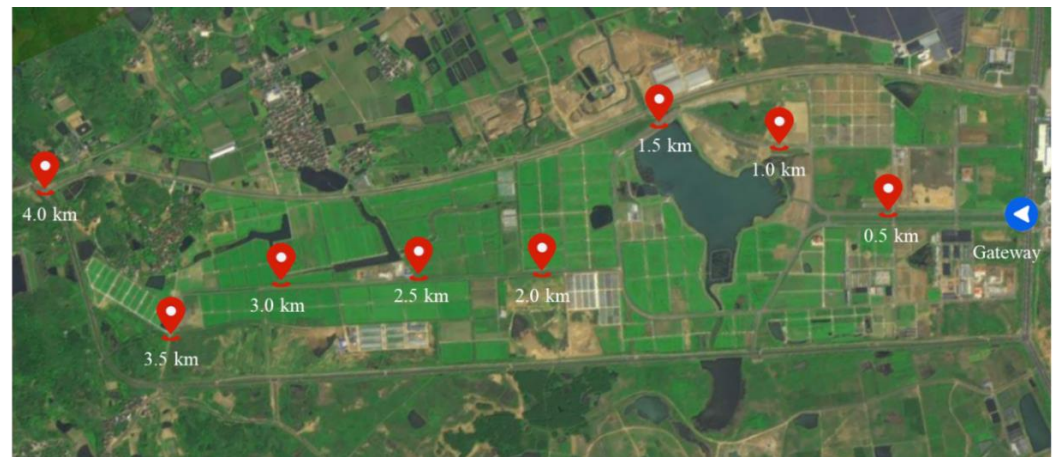


Figure 10. On-site deployment diagram of the communication test gateway and nodes.

The test was conducted under sunny conditions. Figure 11 shows the variation in RSSI and SNR values obtained by the nodes at eight test points at different transmission rates. It can be observed that as the distance increases, the RSSI and SNR values for different rate levels overall show a downward trend. Additionally, there is a significant drop in the RSSI value at the two test points of 1.5 km and 4.0 km. At the 1.5 km point, this is due to the low-lying terrain and proximity to buildings and vehicles, which causes a shadowing multipath effect, blocking some of the radio waves and resulting in higher frequency loss. At the 4.0 km test point, due to the excessive distance reaching the limit of the communication range, the signal is attenuated too much because of path loss, resulting in the lowest received RSSI value; when communicating at higher transmission rates, packet loss occurs. The packet loss rate is 21% at a rate of 10.937 kbps. At a rate of 37.5 kbps, no data packets sent by the gateway can be received. Therefore, without adding a repeater gateway, the maximum communication range of the system is 4 km. At test points within 1.5 km of the gateway, nodes obtained relatively high SNR values when communicating at higher transmission rates. However, as the communication distance increased, nodes obtained higher SNR values when communicating at lower rates. Such changes were because as the distance increased, the signal was subject to more path loss, and communicating at lower rates helped to improve reception sensitivity and anti-interference capabilities.

However, communicating at lower rates reduces the communication efficiency of the system. As shown in Figure 12, which illustrates the transmission time required to send 100 bytes from the gateway to various test points at different transmission rates, when the transmission rate is 10.937 kbps or higher, it takes less than 0.9 s to send 100 bytes to the system's maximum communication range of 4.0 km point. In contrast, communicating at a rate of 0.268 kbps requires up to 6 s for transmission. Therefore, users can deploy the network according to the application scenario requirements based on the above test conditions to ensure signal quality and real-time communication.

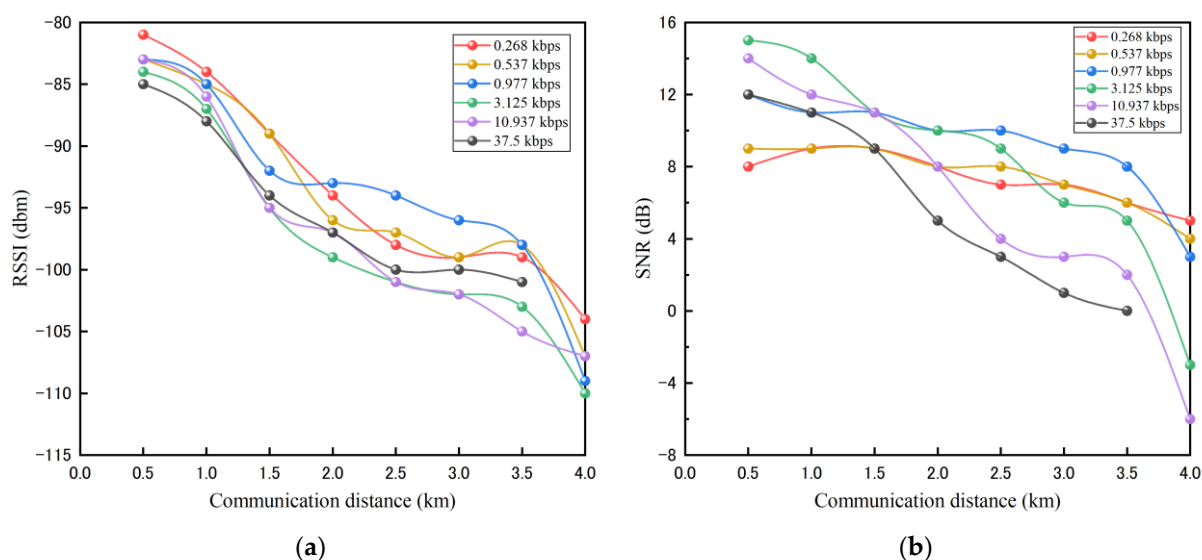


Figure 11. Communication quality. (a) RSSI values; (b) SNR values.

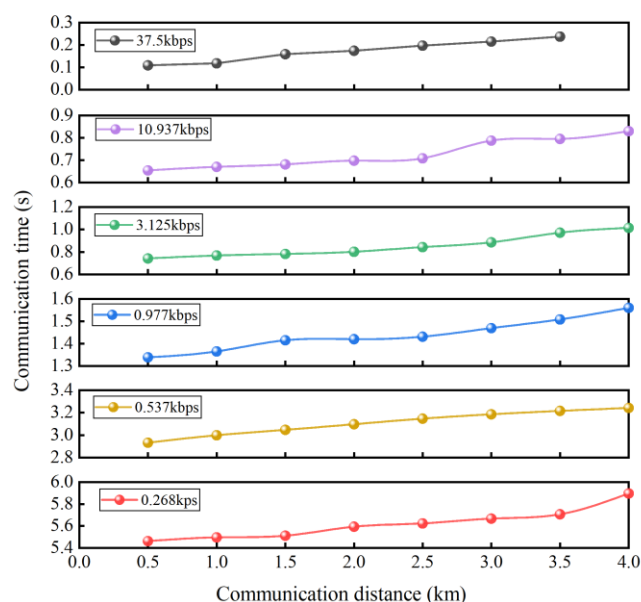


Figure 12. Transmission time at different rates.

3.2. System Irrigation Demonstration

The systematic field irrigation trials were conducted on winter wheat in the grain-filling stage at the Farmland Irrigation Research Institute (FIRI) of the Chinese Academy of Agricultural Sciences (CAAS). The soil bulk density at the trial plots is 1.54 g/cm^3 . The average soil moisture capacity in the field (gravimetric moisture content) for the 0–40 cm soil layer is 17.5%, and for the 0–60 cm soil layer, it is 19.6%. The trials set up two irrigation plots, each 32 square meters (8 m in length and 4 m in width), and used two nodes to monitor the information of the two plots separately. Figure 13 shows the deployment of the gateway and nodes in the field. The nodes and the gateway were powered by solar energy, and the main control circuit boards were placed in metal waterproof boxes. Soil moisture sensors, flow meters, and valves were connected to the node devices, and a small meteorological station was integrated into the gateway.

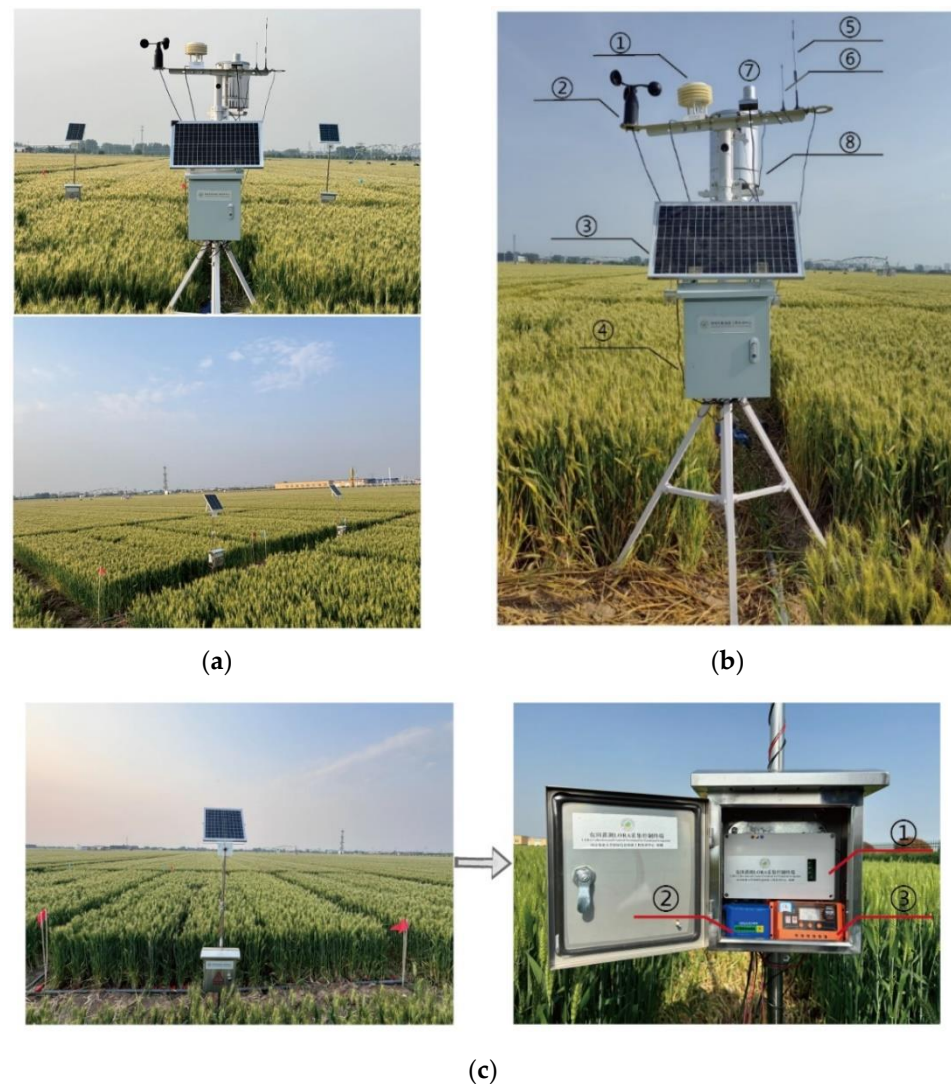


Figure 13. Field deployment diagram. (a) Overview. (b) System gateway and meteorological station: ① instrument shelter (built-in temperature, humidity, and atmospheric pressure sensors); ② wind speed sensor; ③ solar panel; ④ control unit; ⑤ 4G signal gain antenna; ⑥ LoRa signal gain antenna; ⑦ sunshine duration sensor; ⑧ rain gauge. (c) System node: ① control unit; ② 12 V battery pack; ③ solar controller.

The diameter of the main pipe was 1.5 inches, which was connected to the flow meter and valve, as shown in Figure 14a,b. The drip tapes had a diameter of 16 mm and a thickness of 0.4 mm, with a dual-hole spacing of 10 cm, and were equally placed in the field, as shown in Figure 14c. Since the wheat root activity layer could grow up to 60 cm during the grain filling stage [41], a profile with a depth of 60 cm was dug at the center of the field, as shown in Figure 14d. A soil moisture sensor with a 0–50 cm detection depth was placed inside, as shown in Figure 14e, to monitor the moisture content of the 20–60 cm soil layer. According to Table 3, the upper and lower limits of the suitable volumetric moisture content for wheat growth during the early–middle period of the grain-filling stages are 25.6% and 21.5%. Considering the measurement error of $\pm 2\%$ of the sensor, the lower limit for triggering irrigation is set between 21.1% and 21.9%. Moreover, studies have shown that the greatest water consumption occurs within the top 40 cm of soil and that the root system primarily absorbs water from the 20 cm to 40 cm layer. Therefore, the average moisture content at 20 cm, 30 cm, and 40 cm soil depths is used to determine whether there is a water shortage [36].

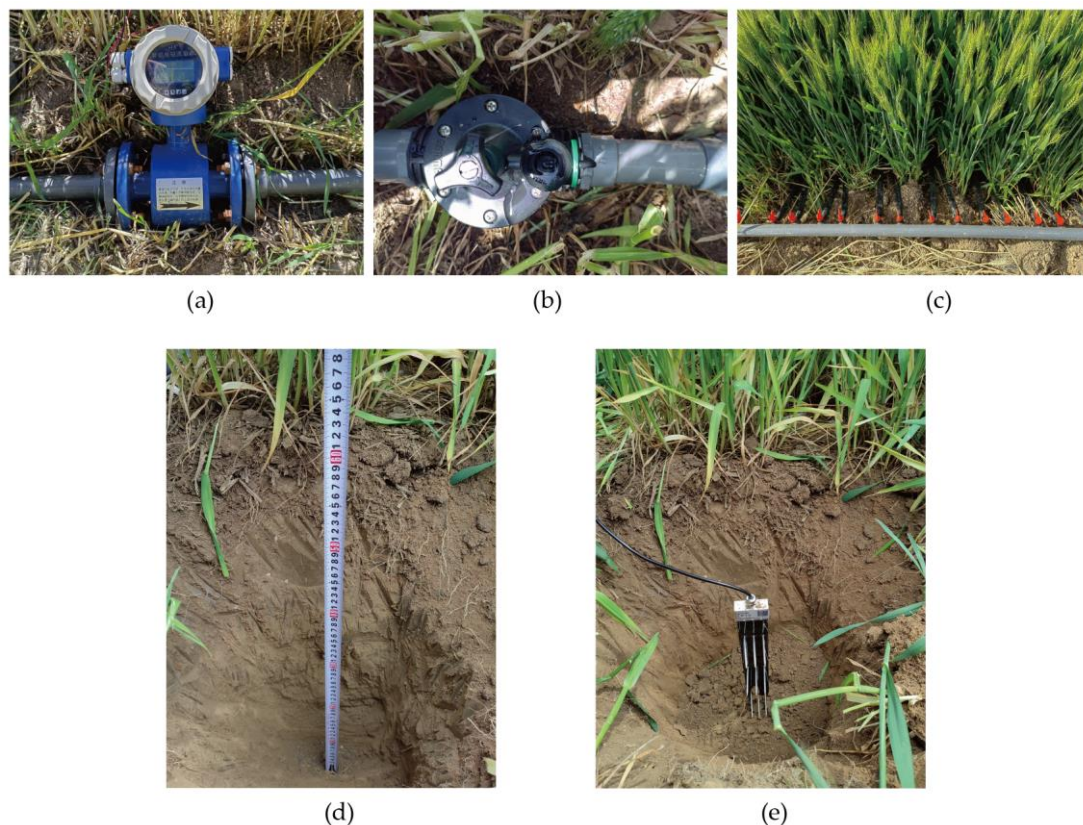


Figure 14. Installation of soil moisture sensors, flow meters, and water valves, along with the laying of pipes and drip irrigation tapes. (a) Magnetic Flow Meter, (b) Pulse Valve, (c) Install drip tapes, (d) 60 cm soil profile, (e) Install soil moisture sensors.

The nodes and gateway acquire soil moisture and meteorological information every three hours, dividing the day into eight periods. At the end of each day, the evapotranspiration is calculated, and a two-day irrigation cycle is used to determine whether irrigation is required. The trials' design involved making irrigation decisions at 9:00 am for plot 1 and at 3:00 pm for plot 2 to explore the impact of different irrigation times on the irrigation system. Additionally, the feasibility of the irrigation system is assessed by comparing the average value obtained after irrigation, when the moisture content of each soil layer has stabilized, with the upper limit soil moisture content set for suitable wheat growth.

The trials started on 26 April 2024 and ended on 16 May 2024, during the critical water demand period of winter wheat, specifically the grain-filling stage. Figure 15 shows the reference evapotranspiration (ET_0) calculated using real-time meteorological data and the modified PM formula. Additionally, with the aid of IBM SPSS Statistics 27 software, we analyzed the correlations between ET_0 and minimum temperature, maximum temperature, mean temperature, sunshine duration, mean wind speed, and mean humidity. The results showed that these meteorological factors have varying degrees of correlation with ET_0 , which is consistent with the conclusions of previous studies by Ma et al. [42]. This further validates the reliability and accuracy of the data collected by the irrigation system. Figure 16 specifically illustrates the degree of correlation between ET_0 and minimum temperature, maximum temperature, mean temperature, sunshine duration, mean wind speed, and mean humidity, with correlation coefficients R^2 of 0.649, 0.864, 0.835, 0.831, 0.582, and -0.702 , respectively. It is evident that the maximum temperature was the most critical factor affecting the water demand during the wheat grain-filling stage in the trial area.

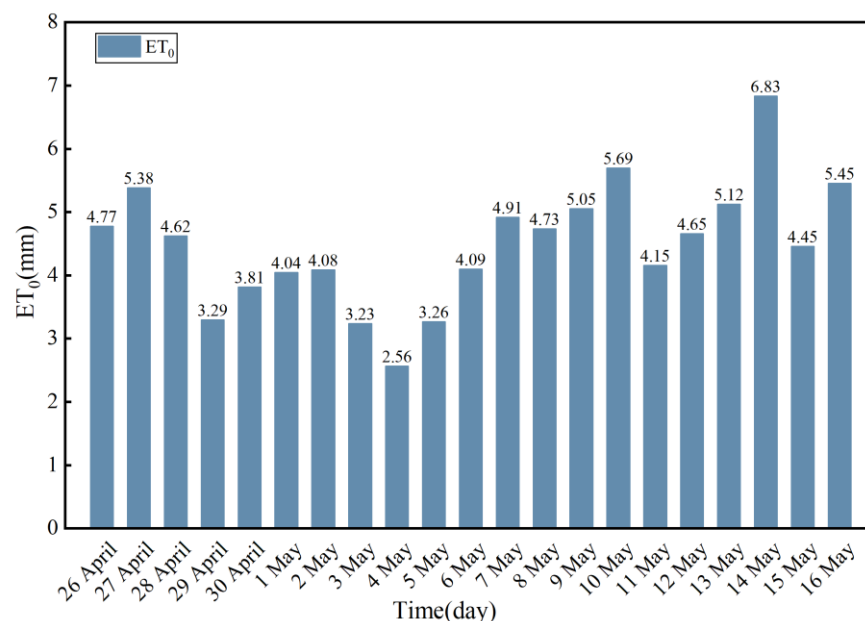


Figure 15. The daily reference evapotranspiration (ET_0) calculated using real-time meteorological data and the modified Penman–Monteith formula.

During the monitoring period, the soil moisture content in both plots only fell below the lower limit of the suitable moisture content twice, and two irrigations were carried out. The irrigation parameters are shown in Table 5. The evapotranspiration (ET_0) and rainfall shown in the table are the total amounts accumulated in the two plots up to the day of irrigation and cleared after each irrigation. The amount of irrigation is calculated based on the irrigation model presented above and is categorized into initial and supplemental irrigation. The first time is classified as initial irrigation, and its irrigation volume is calculated using Equation (13) in combination with the irrigated area. The second irrigation is a supplemental irrigation calculated by combining the irrigated area using Equation (14).

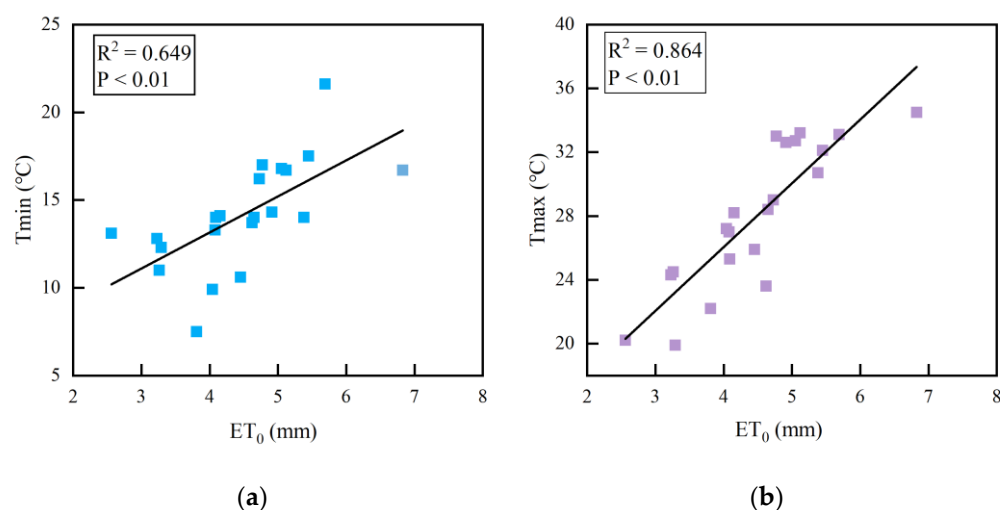


Figure 16. Cont.

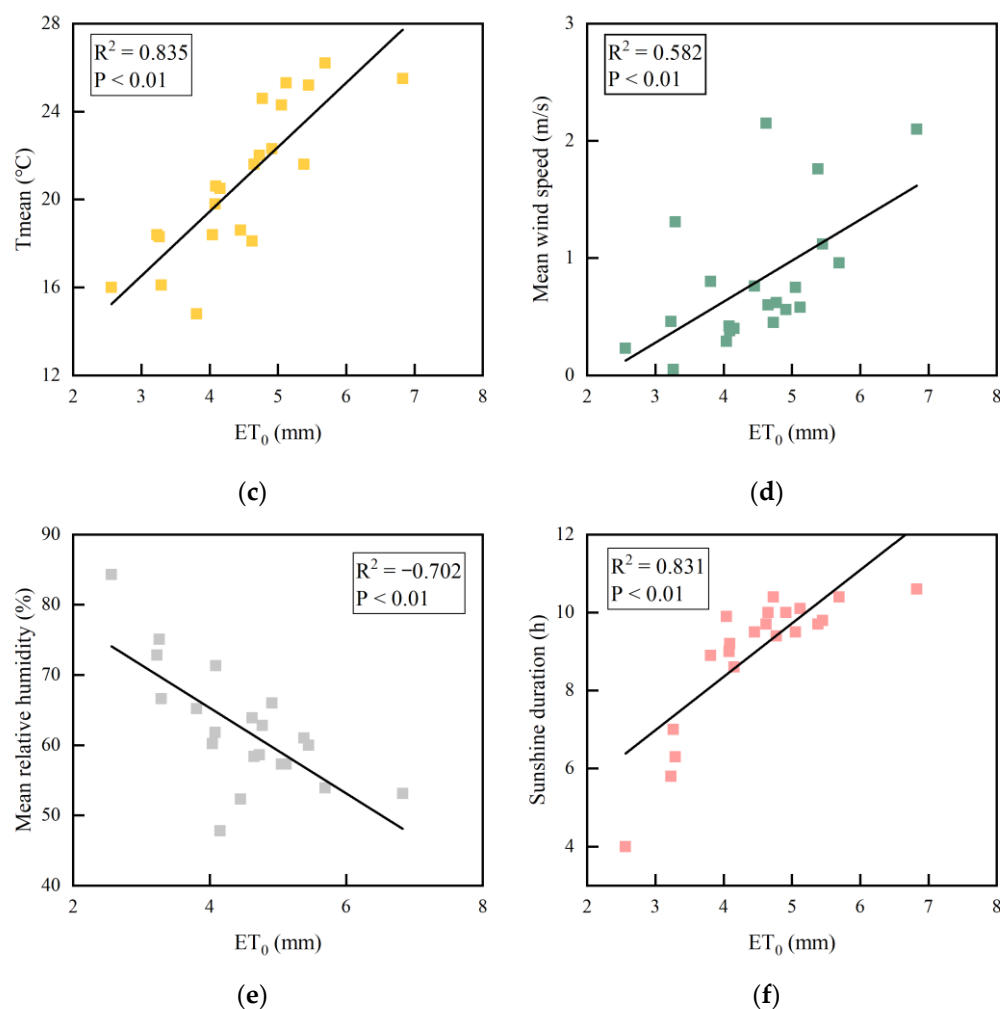


Figure 16. Correlations between meteorological elements and ET_0 . (a) minimum temperature, (b) maximum temperature, (c) mean temperature, (d) sunshine duration, (e) mean wind speed, (f) mean humidity.

Table 5. Irrigation parameters table.

Plot Number	Irrigation Sequence	Date (day)	Cumulative ET_c (mm)	Cumulative Rainfall (mm)	Irrigation Volume (m^3)
Plot 1	First	29 April 2024	5.31	0	1.06
	Second	9 May 2024	30.94	2.7	0.9
Plot 2	First	27 April 2024	5.48	0	1.54
	Second	12 May 2024	42.21	2.7	1.26

Before and after two irrigations of two plots, Figure 17 shows the change curves in the volume moisture content of the 20–60 cm soil layers, which show a sudden change to gradual stabilization. The two reference lines in the figure indicate the moisture content of the soil at the end of irrigation and when the change tends to stabilize. Following the first and second irrigations, Plot 1 average volumetric moisture contents in the 20–60 cm soil layer were 23.96% and 23.38%, respectively, while Plot 2 averages were 25.3% and 23.4%, all within the soil moisture content range suitable for wheat growth. The percentage differences between the average soil moisture content after irrigation and the upper limit of the suitable moisture content of 25.6% in the two plots were -6.4% , -8.68% , -1.12% ,

and -8.59% , respectively, all of which were above 90% of the upper limit, indicating that the irrigation decision system shows good feasibility for different irrigation time points.

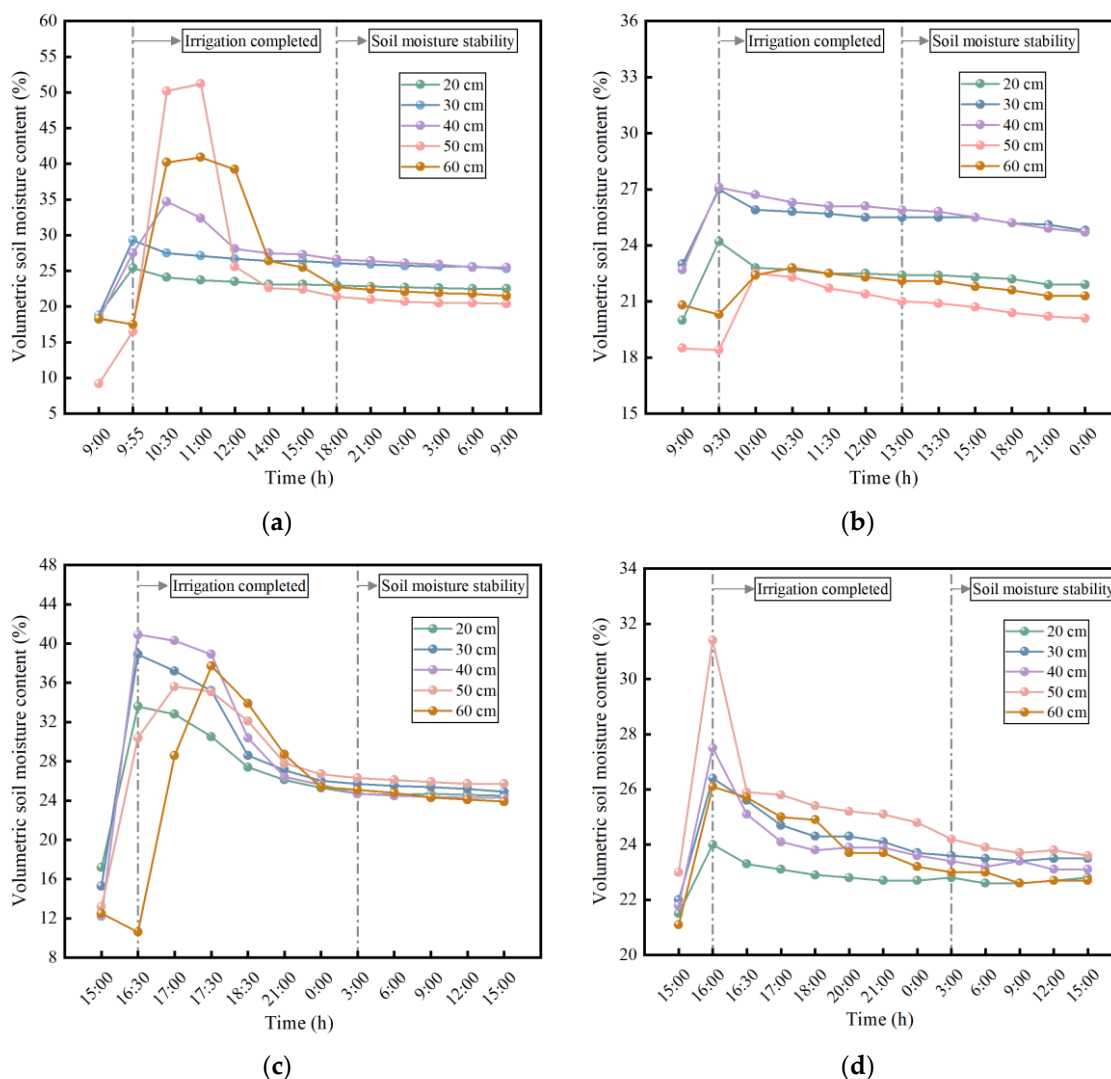


Figure 17. Changes in soil moisture content before and after irrigation. (a) Plot 1 of first irrigation; (b) Plot 1 of second irrigation; (c) Plot 2 of first irrigation; (d) Plot 2 of second irrigation.

Figure 18 shows the curves of the soil moisture content (20–60 cm) in two fields after irrigation as a function of ET_c . The upward trend in the figure is due to the second irrigation. Apart from this, the water content of the soil layer 20–60 cm shows a downward trend with the cumulative increase in ET_c , and the greater the ET_c , the more significantly the soil moisture content decreases. Of these, there is the greater correlation between the change in ET_c and the moisture content of the 40 cm soil layer. Among the meteorological factors, the maximum temperature has the greatest effect on the decrease in soil water content.

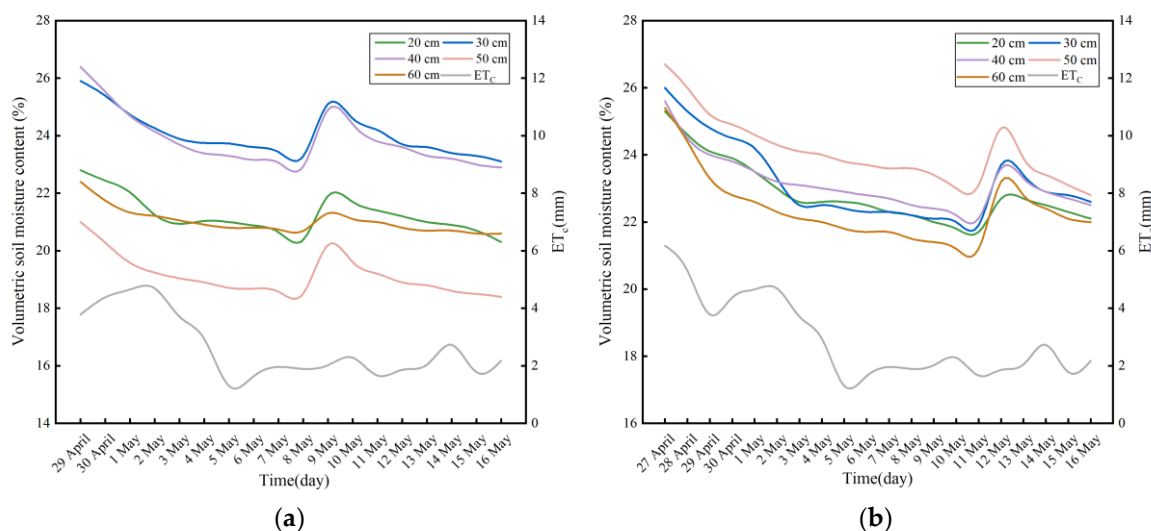


Figure 18. Soil moisture content and ET_c change curve. (a) Plot 1; (b) Plot 2.

4. Conclusions and Discussion

In order to improve the efficiency of irrigation water use and promote the intelligence of agricultural irrigation, it is necessary to develop a suitable smart irrigation system for farmland and efficient water conservation. In this study, an ET-based smart irrigation system for agricultural farmland was developed using LoRa and EC-IoT architecture. The system includes nodes that integrate information sensing and automatic irrigation control, as well as edge gateways that support multiple communication protocols. Irrigation decision-making algorithms were deployed on the gateway to provide localized services for the nodes. Additionally, a WeChat mini program was developed using JavaScript and WeChat developer tools, providing users with interfaces to visualize field data and enabling the remote irrigation control. The main results of the research are as follows.

- (1) The results of the communication test between system nodes and the gateway demonstrate that the constructed LoRa wireless perception network has stable data transmission and strong scalability, as indicated by the maximum communication coverage of 4 km in the absence of a relay gateway, and the packet loss rate of network communication is 0 within 3.5 km. Additionally, the test results found that using a lower transmission rate for communication can achieve longer transmission distances, and the RSSI and SNR values obtained at longer distances are relatively higher but took longer to move. These results can provide users with references for deploying sensing networks.
- (2) For irrigation, an algorithm decision was established using the modified Penman–Monteith formula and water balance equation, allowing for precise irrigation volumes that are adapted to soil characteristics and different stages of crop growth. The field irrigation results during the grain-filling stage of wheat indicate that the soil moisture content after irrigation was within the suitable moisture range and above 90% of the suitable upper limit value, showing that the irrigation system is performing effectively and has initially met the objective of precision irrigation.

This study has preliminarily applied the edge computing architecture to the scenario of agricultural irrigation. By forming a LoRa wireless sensor network and constructing a decision model for wheat irrigation, the system has achieved the collection of soil moisture and environmental information, remote monitoring, and automated control of the irrigation process. According to the test results, it is shown that the system can ensure stable data transmission, meet the needs of monitoring extensive farmland areas, and the irrigation

effect is good, providing a reference for the irrigation management of wheat. However, the smart irrigation system in this study is currently applied to wheat crops. In future research, we intend to conduct more trials on different crops with different fertility periods to continuously improve the irrigation decision algorithms. This study has preliminarily realized the migration of the irrigation algorithm from the cloud to the gateway for irrigation management. Subsequent research can add firmware update functionality to the gateway and establish a cloud-based gateway management system, enabling edge gateways to conveniently obtain the latest irrigation algorithms and other functions, thereby enhancing their flexibility and scalability. Furthermore, we can integrate meteorological forecast information into the algorithm and use technologies such as deep learning to predict irrigation amounts and optimize irrigation schedules, thereby enhancing the system's adaptability under complex weather conditions.

Author Contributions: Y.Z. (Ying Zhang), X.W. and X.J. developed the IoT irrigation system; X.J., Y.Z. (Yan Zhu), W.C., J.N. and L.J. conceived and designed the experiments; Y.Z. (Ying Zhang) and X.W. performed the experiments; Y.Z. (Ying Zhang), X.J. and X.W. analyzed the data; Y.Z. (Ying Zhang) wrote the paper. All authors have read and agreed to the published version of the manuscript.

Funding: This work was supported by the Primary Research and Development Plan of Jiangsu Province (grant number BE2021308, BE2019306, BE2024008).

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The dataset used in this research is available upon valid request to any of the authors of this research article.

Acknowledgments: The authors gratefully acknowledge the financial support provided by the Primary Research and Development Plan of Jiangsu Province (grant numbers BE2021308, BE2019306, BE2024008). This work was supported by Jiangsu Rainfolks Irrigation System Co., Ltd., the National Information Agricultural Engineering Technology Center, the Engineering Research Center of Smart Agriculture, Ministry of Education, and the Jiangsu Collaborative Innovation Center for the Technology and Application of Internet of Things. Furthermore, the authors appreciate Xiaoping Jiang, Jun Ni, Yan Zhu, and Weixing Cao for their fruitful discussions. Finally, the authors are grateful to four anonymous reviewers for their valuable comments.

Conflicts of Interest: Author Liyong Jin was employed by the company Jiangsu Rainfolks Irrigation System Co., Ltd. The remaining authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

References

1. Sustainable Development Goals. Goal 2: Zero Hunger. Available online: <https://www.un.org/sustainabledevelopment/hunger> (accessed on 10 October 2024).
2. United Nations. UN Projects World Population to Peak Within This Century. Available online: <https://www.un.org/en/UN-projects-world-population-to-peak-within-this-century> (accessed on 10 October 2024).
3. UN World Water Development Report. Food Security, a Key Driver of Peace and Prosperity. Available online: <https://www.unesco.org/reports/wwdr/en/2024/agriculture?hub=2> (accessed on 10 October 2024).
4. Gao, H.R.; Zhangzhong, L.L.; Zheng, W.A.; Chen, G.F. How can agricultural water production be promoted? a review on machine learning for irrigation. *J. Clean. Prod.* **2023**, *414*, 137687. [CrossRef]
5. Duguma, A.L.; Bai, X. Contribution of Internet of Things (IoT) in improving agricultural systems. *Int. J. Environ. Sci. Technol.* **2024**, *21*, 2195–2208. [CrossRef]
6. Obaideen, K.; Yousef, B.A.A.; Almallahi, M.N.; Tan, Y.C.; Mahmoud, M.; Jaber, H.; Ramadan, M. An overview of smart irrigation systems using IoT. *Energy Nexus* **2022**, *7*, 100124. [CrossRef]

7. Froiz-Míguez, I.; Lopez-Iturri, P.; Fraga-Lamas, P.; Celaya-Echarri, M.; Blanco-Novoa, O.; Azpilicueta, L.; Falcone, F.; Fernández-Caramés, T.M. Design, Implementation, and Empirical Validation of an IoT Smart Irrigation System for Fog Computing Applications Based on LoRa and LoRaWAN Sensor Nodes. *Sensors* **2020**, *20*, 6865. [\[CrossRef\]](#) [\[PubMed\]](#)
8. Tang, P.; Liang, Q.; Li, H.; Pang, Y.Y. Application of Internet-of-Things Wireless Communication Technology in Agricultural Irrigation Management: A Review. *Sustainability* **2024**, *16*, 3575. [\[CrossRef\]](#)
9. Ferrández-Pastor, F.J.; García-Chamizo, J.M.; Nieto-Hidalgo, M.; Mora-Martínez, J. Precision Agriculture Design Method Using a Distributed Computing Architecture on Internet of Things Context. *Sensors* **2018**, *18*, 1731. [\[CrossRef\]](#)
10. Gutiérrez, J.; Villa-Medina, J.F.; Nieto-Garibay, A.; Porta-Gándara, M.A. Automated Irrigation System Using a Wireless Sensor Network and GPRS Module. *IEEE Trans. Instrum. Meas.* **2014**, *63*, 166–176. [\[CrossRef\]](#)
11. Goap, A.; Sharma, D.; Shukla, A.K.; Krishna, C.R. An IoT based smart irrigation management system using Machine learning and open source technologies. *Comput. Electron. Agric.* **2018**, *155*, 41–49. [\[CrossRef\]](#)
12. Ramson, S.R.J.; León-Salas, W.D.; Brecheisen, Z.; Foster, E.J.; Johnston, C.T.; Schulze, D.G.; Filley, T.; Rahimi, R.; Soto, M.; Bolivar, J.A.L.; et al. A Self-Powered, Real-Time, LoRaWAN IoT-Based Soil Health Monitoring System. *IEEE Internet Things J.* **2021**, *8*, 9278–9293. [\[CrossRef\]](#)
13. Liopa-Tsakalidi, A.; Thomopoulos, V.; Barouchas, P.; Kavga, A.; Boursianis, A.D.; Goudos, S.K.; Kalamaras, G.; Gotsis, A.; Maliatsos, K. A NB-IoT based platform for smart irrigation in vineyard. In Proceedings of the 10th International Conference on Modern Circuits and Systems Technologies (MOCAS), Thessaloniki, Greece, 5–7 July 2021.
14. Almuhaaya, M.A.M.; Jabbar, W.A.; Sulaiman, N.; Abdulmalek, S. A Survey on LoRaWAN Technology: Recent Trends, Opportunities, Simulation Tools and Future Directions. *Electronics* **2022**, *11*, 164. [\[CrossRef\]](#)
15. Qin, J.J.; Wang, L.P.; Li, H.T. Frost Prevention and Mitigation System of Mountain Orchard Based on LoRa. In Proceedings of the 17th IEEE International Wireless Communications and Mobile Computing Conference (IEEE IWCMC), Harbin City, China, 28 June–2 July 2021.
16. Nyamuryekung'e, S.; Duff, G.; Utsumi, S.; Estell, R.; McIntosh, M.M.; Funk, M.; Cox, A.; Cao, H.P.; Spiegel, S.; Perea, A.; et al. Real-Time Monitoring of Grazing Cattle Using LORA-WAN Sensors to Improve Precision in Detecting Animal Welfare Implications via Daily Distance Walked Metrics. *Animals* **2023**, *13*, 2641. [\[CrossRef\]](#) [\[PubMed\]](#)
17. Sendra, S.; Parra, L.; Jimenez, J.M.; Garcia, L.; Lloret, J. LoRa-based Network for Water Quality Monitoring in Coastal Areas. *Mob. Netw. Appl.* **2023**, *28*, 65–81. [\[CrossRef\]](#)
18. Benzaouia, M.; Hajji, B.; Mellit, A.; Rabhi, A. Fuzzy-IoT smart irrigation system for precision scheduling and monitoring. *Comput. Electron. Agric.* **2023**, *215*, 108407. [\[CrossRef\]](#)
19. Peng, Y.Q.; Xiao, Y.X.; Fu, Z.T.; Dong, Y.H.; Zheng, Y.J.; Yan, H.J.; Li, X.X. Precision irrigation perspectives on the sustainable water-saving of field crop production in China: Water demand prediction and irrigation scheme optimization. *J. Clean. Prod.* **2019**, *230*, 365–377. [\[CrossRef\]](#)
20. Vianny, D.M.M.; John, A.; Mohan, S.K.; Sarlan, A.; Ahmadian, A. Water optimization technique for precision irrigation system using IoT and machine learning. *Sustain. Energy Technol. Assess.* **2022**, *52*, 102307. [\[CrossRef\]](#)
21. Preite, L.; Vignali, G. Artificial intelligence to optimize water consumption in agriculture: A predictive algorithm-based irrigation management system. *Comput. Electron. Agric.* **2024**, *223*, 109126. [\[CrossRef\]](#)
22. Allen, R.G.; Pereira, L.S.; Raes, D.; Smith, M. *Crop Evapotranspiration-Guidelines for Computing Crop Water Requirements*; Food and Agriculture Organization of the United Nations: Rome, Italy, 1998; pp. 15–134.
23. Davis, S.L.; Dukes, M.D. Irrigation scheduling performance by evapotranspiration-based controllers. *Agric. Water Manag.* **2010**, *98*, 19–28. [\[CrossRef\]](#)
24. Dela Cruz, J.C.; Caya, M.V.C.; Ballado, A.H. Evapotranspiration-based Irrigation System for Mustard Green Crop Cultivation using Public Weather Forecast. In Proceedings of the 11th IEEE Control and System Graduate Research Colloquium (ICSGRC), Shah Alam, Malaysia, 8 August 2020; pp. 203–208.
25. Caya, M.V.; dela Cruz, J.; Merina, J.P.M.; Mora, D.K.H.; Bentir, S.A.P. Development of ET-Based Irrigation System in Green Roofs Using Penman-Monteith Equation. In Proceedings of the 11th IEEE International Conference on Humanoid, Nanotechnology, Information Technology, Communication and Control, Environment, and Management (HNICEM), Laoag, Philippines, 29 November–1 December 2019.
26. Lopez, P.G.; Montesor, A.; Epema, D.; Datta, A.; Higashino, T.; Iamnitshi, A.; Barcellos, M.; Felber, P.; Riviere, E. Edge-centric Computing: Vision and Challenges. *ACM SIGCOMM Comput. Commun. Rev.* **2015**, *45*, 37–42. [\[CrossRef\]](#)
27. Gope, P.; Lee, J.; Hsu, R.H.; Quek, T.Q.S. Anonymous Communications for Secure Device-to-Device Aided Fog Computing. *IEEE Consum. Electron. Mag.* **2019**, *8*, 10–16. [\[CrossRef\]](#)
28. Jiang, C.F.; Fan, T.T.; Gao, H.H.; Shi, W.S.; Liu, L.K.; Cérin, C.; Wan, J. Energy aware edge computing: A survey. *Comput. Commun.* **2020**, *151*, 556–580. [\[CrossRef\]](#)

29. Schneider Electric, Cost-Benefit Analysis of Deploying Edge Micro Data Centers White Paper. Available online: https://download.schneider-electric.com/files?p_Doc_Ref=SPD_VAVR-9X6SVK_CH&p_enDocType=White+Paper&p_File_Name=VAVR-9X6SVK_R0_CH.pdf (accessed on 10 October 2024).
30. Fan, Q.; Li, Z.; Chen, X. Inference delay optimization of branchy neural network model based on edge computing. *Comput. Appl.* **2020**, *40*, 342–346.
31. Huang, C.L.; Ke, Y.X.; Hua, X.D.; Yang, J.Y.; Sun, M.Y.; Yang, W. Neng Current status and prospects of edge computing in smart agriculture. *Trans. Chin. Soc. Agric. Eng.* **2022**, *38*, 224–234.
32. Yang, J.; Li, G.Q.; Ma, J.Y.; Lei, Y.G. Design and implementation of a Smart Internet IoT agricultural pest identification system based on edge computing. *Henan Sci. Technol.* **2024**, *51*, 32–37. [[CrossRef](#)]
33. Zhang, Q.C.; Wang, J.Y.; Lu, Y.H.; Wang, W. Design of Control System for ARM Vehicle Robot. In Proceedings of the International Conference on Robots and Intelligent System (ICRIS), Haikou, China, 15–16 June 2019; pp. 22–25.
34. da Silva, L.M.; Menezes, H.B.D.; Luccas, M.D.; Mailer, C.; Pinto, A.S.R.; Boava, A.; Rodrigues, M.; Ferrao, I.G.; Estrella, J.C.; Branco, K. Development of an Efficiency Platform Based on MQTT for UAV Controlling and DoS Attack Detection. *Sensors* **2022**, *22*, 6567. [[CrossRef](#)] [[PubMed](#)]
35. Wang, W.J.; Feng, H. Research on the water requirement patterns and irrigation systems of winter wheat in the Guanzhong region based on CROPWAT-DSSAT. *Chin. J. Eco-Agric.* **2012**, *20*, 795–802. [[CrossRef](#)]
36. Chen, Y.M.; Guo, G. *Shuang: Water Requirements and Irrigation of Major Crops in China*; China Water & Power Press: Beijing, China, 1995; pp. 175–199.
37. Afzaal, H.; Farooque, A.A.; Abbas, F.; Acharya, B.; Esau, T. Precision Irrigation Strategies for Sustainable Water Budgeting of Potato Crop in Prince Edward Island. *Sustainability* **2020**, *12*, 2419. [[CrossRef](#)]
38. Tong, C.L.; Zhang, W.J.; Tang, Y.; Wang, H.Q. Daily solar radiation simulation and calculation. *Chin. J. Agrometeorol.* **2005**, *26*, 5.
39. Zuo, D.K.; Wang, Y.X.; Chen, J.S. Spatial distribution characteristics of total solar radiation in China. *Acta Meteorol. Sin.* **1963**, *1*, 78–96.
40. Cui, D.D.; Zhang, Q.; Tian, X.X.; Han, Z.; Liu, P.Z. Irrigation decision management for ‘Shine Muscat’ grapes based on the water balance equation. *Sino-Overseas Grapevine Wine* **2022**, *4*, 23–29. [[CrossRef](#)]
41. Li, Y.; Ma, J.Q.; Zhao, X.S. Real-time Soil Moisture Content and Root Zone Analysis of Planned Wet Layer for Winter Wheat. *Water Sav. Irrig.* **2013**, 14–17.
42. Ma, Y.L.; Niu, Z.R.; Sun, D.Y. Spatial and Temporal Patterns of Potential Evapotranspiration and Their Relationship with Meteorological Factors in the Hexi Corridor. *Arid. Land Geogr.* **2024**, *47*, 192–202.

Disclaimer/Publisher’s Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.